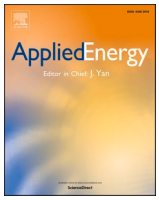
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Decarbonization dispatch for AI computing center parks considering task-level carbon traceability☆

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HIGHLIGHTS

- Task-level carbon tracing is embedded in AI park decarbonization dispatch.
- Three linked ledgers close task, energy, and carbon-attribute accounting.
- Storage preserves clean-electricity provenance and prevents duplicate claims.
- CVaR constraints control task carbon-label risks under uncertainty.
- Park tests cut operating cost by 6.6% and carbon emissions by 11.3%.

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ABSTRACT

The decarbonization dispatch of AI computing centers should not stop at controlling total park-level carbon emissions; it should also trace the energy sources, carbon-emission responsibility, and low-carbon labels of individual AI tasks. To address the limitation that existing studies mainly adopt park-average accounting and therefore cannot support task-level accounting, this paper proposes a decarbonization dispatch optimization method for AI computing center parks considering task-level carbon traceability. Taking the task execution trajectory as the main line, the proposed method links resource-pool startup and shutdown, IT power consumption, thermally inertial cooling, network storage, and energy-storage operation, thereby forming a closed relationship among the task ledger, energy-consumption ledger, and low-carbon ledger. Meanwhile, a unique low-carbon electricity claim and a label-violation CVaR constraint are introduced to control the risk of label failure under uncertain scenarios. Rolling-dispatch results on the Chenggong, Yunnan, China and the modified IEEE 118-bus system show that the proposed method can significantly improve task-level low-carbon performance with only a small increase in low-carbon attribute procurement and energy-storage throughput. In the Chenggong park case, the operating cost is reduced by 6.6%, and market-based park carbon emissions are reduced by 11.3%. The method provides a dispatch modeling basis for shifting low-carbon computing services from park-average accounting to task-level verification.

1. Introduction

Large-model training, batch inference, and multimodal services are rapidly increasing the energy, water, and carbon footprint of AI server infrastructure [1]. De Vries-Gao [2] notes that operators' environmental reports often do not distinguish AI from non-AI workloads, making AI-specific carbon and water footprints difficult to quantify from enterprise-level metrics. Greenhouse-gas inventory research shows that efficiency measures and renewable electricity can substantially reduce data-center emissions, while net-zero operation may still require mitigation and compensation across the value chain [3]. These enterprise-level accounting approaches do not by themselves provide a traceability mechanism for an individual computing task. The 24/7 carbon-free electricity literature makes the point from the supply side—clean computing should be matched to low-carbon electricity in time,

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Nomenclature

Abbreviations

AIDC	Artificial Intelligence Data Center
AI	Artificial Intelligence
IT	Information Technology
SLA	Service Level Agreement
CVaR	Conditional Value-at-Risk
COP	Coefficient of Performance
PUE	Power Usage Effectiveness
PCS	Power Conversion System
PDU	Power Distribution Unit
PPA	Power Purchase Agreement
REC	Renewable Energy Certificate
SOC	State of Charge
PV	Photovoltaic
MIP	Mixed-Integer Programming
DVFS	Dynamic Voltage and Frequency Scaling
DAG	Directed Acyclic Graph

Sets and indices

$T=\{1,2,\ldots,N_T\}$	Set of dispatch periods, indexed by t
$T^{+}=\{1,2,\ldots,N_{T+1}\}$	Set of state periods
$J=\{1,2,\ldots,N_J\}$	Set of AI tasks, indexed by j
J^{low}	Set of tasks that apply for or purchase low-carbon computing labels
$G=\{1,2,\ldots,N_G\}$	Set of computing resource pools, indexed by g
$Z=\{1,2,\ldots,N_Z\}$	Set of cooling zones, indexed by z
$S=\{1,2,\ldots,N_S\}$	Set of uncertainty scenarios, indexed by s
l	Lag-order index of thermal response
$z(g)$	Cooling zone to which resource pool g belongs

Parameters

d_t	Length of a single dispatch period
a_j, d_j, W_j	Arrival time, deadline, and computational workload of task j
L_j^{ckpt}	Progress loss of task j when checkpoint recovery occurs
R_g	Maximum equivalent computing-service rate that resource pool g can provide in a single period
P_g^{idle}	Base power of resource pool g when it is on
$k_{j,g}^{it}$	Unit computing-service power coefficient of task j on resource pool g
$P_t^{aux,base}$	Base auxiliary power in period t
$E_{0,t}^{net}$	Measured or forecast energy of background network, storage, and auxiliary systems in period t
$\overline{P}_{t,s}^{ren}$	Available renewable generation power in period t under scenario s
$P^{-\mathrm{grid}}$	Upper limit of the park grid-connection capacity
$P^{-\mathrm{ch}}$, $P^{-\mathrm{dis}}$	Maximum charging and discharging power of the energy storage system
S^{-}	Upper limit of energy storage capacity
S^{ini}, S^{ter}	Initial and terminal states of total stored energy
$S^{\mathrm{low,ini}}$, $S^{\mathrm{low,ter}}$	Initial and terminal states of low-carbon energy inventory in storage
n^{ch}, n^{dis}	Charging and discharging efficiencies of the energy storage system
$\overline{E}_{t,s}^{low,grid}$	Upper limit of externally purchased low-carbon electricity that can be purchased or claimed in period t under scenario s
$m_{z,l}^{th}$	lth-order thermal response kernel coefficient of cooling zone z
H_z	Maximum thermal response lag order of cooling zone z
M_z^{th}	Sum of thermal response kernel coefficients of cooling zone z
$COP_{z,t,s}$	Coefficient of performance of cooling zone z in period t under scenario s
k_g^{heat}	Proportional coefficient converting the electrical power of resource pool g into heat load
a_j^{io}, a_j^{sync}, a_j^{ckpt}	Unit energy coefficients for task data read/write, parameter synchronization, and checkpoint read/write
k_j^{sync}	Proportional coefficient between the parameter synchronization volume and computational workload of task j
$\overline{D}_j^{io}$, $\overline{D}_j^{sync}$	Upper limits of data read/write volume and parameter synchronization volume of task j in a single period
$\overline{b}_{j,t}^{sto}$	Fixed allocation coefficient of storage losses for task j in period t
$f_{j,g,t}^{iddle}$	Task-level traceability allocation coefficient for resource-pool base energy consumption
D_j^{load}, $\overline{D}_j^{save}$	Data volume loaded at startup and saved to storage at shutdown for task j
k_j^{io}	Data read/write coefficient per unit computational workload of task j
M_t^{sto}	Big-M upper bound used in the linearization of storage-loss allocation
$Q_{z,t}^{env}$	Base heat load from the environment and enclosure structure of cooling zone z
$Q_{z,t}^{aux,heat}$	Equivalent heat load entering cooling zone z from network, storage, power-distribution, and other auxiliary devices
$Q_{z,t}^{hist}$	Thermal-inertia tail term in period t formed by historical heat loads before the beginning of the dispatch window
$\overline{P}_z^{cool}$	Upper limit of cooling capacity of cooling zone z
$EF_{t,s}^{grid}$	Grid emission factor used for location-based physical emission accounting
c_j^{ckpt}, $c^{\mathrm{low,ter}}$	Checkpoint action cost and penalty coefficient for terminal deviation of low-carbon inventory
$EF_{t,s}^{low}$, $EF_{t,s}^{res}$	Low-carbon claim emission factor and residual-mix emission factor in period t under scenario s
$\overline{CI}_j$	Committed upper limit of carbon intensity for low-carbon task j
B_1, B_2, B_3	Grading thresholds of task carbon-intensity labels
$c_{t,s}^{grid}$, $c_{t,s}^{low,grid}$, c^{sto}, c^{cur}, c^{cool}	Cost coefficients for grid electricity purchase, low-carbon attribute premium or REC/PPA price difference, storage cycling, renewable curtailment, and cooling-capacity slack
c^{park}, c^{tag}, c^{SLA}, c^{risk}	Weights of park-level carbon emissions, label violation, SLA shortfall, and tail risk
w_j^{SLA}, w_j^{tag}	SLA shortfall weight and label-violation weight of task j
p_s	Probability of scenario s
α^{cv}	CVaR confidence level for label-violation risk
$R^{-\mathrm{tag}}$	Upper limit of tail violation risk for labels

Variables

$u_{j,g,t}$	Equivalent computing-service rate obtained by task j from resource pool g in period t
$x_{j,g,t}$	Binary variable indicating whether task j occupies resource pool g
$b_{j,t}$	Binary variable indicating whether task j runs in period t
$y_{g,t}$	Binary variable indicating whether resource pool g is on in period t
$n_{j,t}^{sta}$, $n_{j,t}^{stop}$	Startup and shutdown variables of task j in period t
$h_{j,t}^{hist}$	Historical indicator of whether task j has been started before period t
$q_{j,t}^{rst}$, $q_{j,g,g',t}^{tr}$, $q_{j,t}^{mig}$, $q_{j,t}^{ckpt}$	Trigger variables for restart after interruption, resource-pool transition, migration, and checkpoint loss
$r_{j,t}$	Cumulative execution progress of task j up to period t

s_j^{SLA} SLA shortfall variable of task j

$P_{t,s}^{grid}, P_{t,s}^{ren,use}, P_{t,s}^{cur}$ Grid purchase, renewable utilization, and renewable curtailment power under scenario s

$P_{ch}\ t,s$, $P_{dis}\ t,s$ Charging and discharging power of the energy storage system under scenario s

$P_t^{IT}, P_{t,s}^{cool}, P_t^{aux}$ IT, cooling, and auxiliary power

$P_{z,t,s}^{cool}$ Cooling power of cooling zone z

$q_{j,t}^{rst}, q_{j,g,g',t}^{tr}, q_{j,t}^{mig}, q_{j,t}^{ckpt}$ Trigger variables for restart after interruption, resource-pool transition, migration, and checkpoint loss

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$P_t^{IT}, P_{t,s}^{cool}, P_t^{aux}$ IT, cooling, and auxiliary power

$P_{z,t,s}^{cool}$ Cooling power of cooling zone z

$Q_{j,z,t}^{heat}, Q_{0,z,t}^{heat}, \tilde{Q}_{z,t}^{heat}$ Task heat load, background heat load, and total response heat load

$D_{j,t}^{io}, D_{j,t}^{sync}$ Data read/write volume and parameter synchronization volume of a task

$E_{j,g,t}^{dyn}, E_{j,g,t}^{idle}$ Task-level dynamic IT energy and allocated base IT energy

$E_{t,s}^{sto,loss}, E_{t,s}^{meter}$ Equivalent storage loss energy and energy at the park metering boundary

$S_{t,s}$, $S_{t,s}^{low}$ Total stored energy state and low-carbon energy inventory in storage

$z_{t,s}^{ch}, z_{t,s}^{dis}$ Binary charging and discharging status variables of the energy storage system

$E_{j,t,s}^{trace}$ Traceable energy of task j in period t under scenario s

$E_{j,t}^{IT}, E_{j,s,t}^{cool}, E_{j,t}^{net}, E_{j,s,t}^{sto}$ Task-level allocated energy of IT, cooling, network storage, and storage losses

$E_{0,t,s}^{trace}$ Background or unallocated traceable energy

$E_{t,s}^{low,grid}, E_{t,s}^{low,in}, E_{t,s}^{low,ch}, E_{low,dis,t,s}, E_{t,s}^{low,avl}$ Externally purchased low-carbon electricity, low-carbon electricity inflow, charging into storage, release from storage, and claimable electricity

$A_{j,t,s}^{low}, A_{0,t,s}^{low}$ Claimable low-carbon electricity allocated to tasks and background loads

$C_{j,s}^{trace}$ Market-based traceable carbon emissions assigned to task j under scenario s

$C_{0,s}^{trace}$ Market-based carbon emissions associated with background or unallocated electricity consumption

C_s^{prak} Total market-based carbon emissions within the park metering boundary

C_s^{phy} Location-based emissions calculated from physical grid purchases and the grid-average emission factor

$CI_{j,s}$ Carbon intensity of task j under scenario s

$v_{j,s}^{tag}$ Label-violation slack variable of task j under scenario s

V_s^{tag} Label-violation loss under scenario s

$R^{tag}, \xi^{tag}, v_s^{tail}$ Label CVaR risk value, auxiliary quantile variable, and tail auxiliary variable

$v_{z,t,s}^{cool}, v_s^{low,ter}$ Cooling-capacity slack and terminal low-carbon inventory slack variables

F, F_s^{rec} Total objective value and scenario recourse operating cost

location, and consumption trajectory rather than netted out by annual certificates [4]. The question addressed in this paper is therefore not only how much a park's total emissions can be reduced, but whether the carbon claim attached to a single AI task can be made traceable and auditable within the dispatch model itself.

The most direct precedent is carbon-aware computing. Radovanovic et al. [5] demonstrated at production scale that delaying flexible workloads toward hours of low grid carbon intensity reduces operational emissions without materially degrading intraday completion, and later work extended the idea to geographic load balancing under on-site renewables [6]. A useful corrective came from machine-learning assessments of how much flexibility actually remains once cooling mode and server thermal state are accounted for [7]: the exploitable margin is narrower than idealized models suggest. On the task side, realism has improved as well—online scheduling with job dependence, heterogeneous tasks, and service-quality requirements has been cast as a Markov decision process [8], and flexibility assessment from real production logs has shown that ignoring arrival patterns and power mappings overstates the regulation potential of computing loads [9]. What this line of work consistently does not do, however, is close the loop at the level of an individual task. Energy and emissions are reported for a cluster or a data center as a whole, and the link between one task's execution trajectory and a specific, non-reusable quantity of low-carbon electricity is never established.

A second body of work couples the data center with its surrounding energy system. Task migration shifts the operating points of electrical, cooling, and heating equipment simultaneously, so computing-side and energy-side models must be built within one boundary [10]; zero-carbon and renewable-rich configurations have accordingly been studied as multi-energy portfolios, robust operation problems under on-site renewable uncertainty, and spatio-temporal scheduling problems [11–13]. Two findings recur across these studies. First, cost, energy, and carbon do not move together: task transfer that lowers operating cost can raise average delay [14], and cutting energy use is not equivalent to cutting emissions when electricity carbon footprints differ across locations and hours [15,16]. Second, capacity and supply strategies must track grid carbon intensity dynamically rather than being sized to a fixed peak [17]. These works substantially widen the energy boundary of low-carbon data-center operation, but their accounting object remains the park, the data center, or the server cluster; none of them can underwrite a task-level low-carbon claim.

For AI loads specifically, the cooling side is the hard part of task-level traceability. Combined computing–cooling energy models for micro data centers [18] and hierarchical load models that separate interactive from delay-tolerant batch jobs [19] both show that coarse linear load models miss the dynamics that matter for high-precision dispatch; workload migration additionally introduces cross-network coordination constraints [20] and multi-source uncertainty [21]. More important for this paper, cooling efficiency is not a fixed parameter. The spatial placement of workload changes zonal cooling consumption and thermal-safety states [22], and computing-side savings can be offset at the facility side when thermal constraints are neglected [23]. Reviews of reinforcement learning for data-center efficiency and of cooling technology point in the same direction: coordination across subsystems, cooling close to the heat source, and explicit treatment of thermal dynamics [24,25]. The implication is that a credible task-level label cannot be produced by dividing park energy through an average PUE or park-average carbon factor; it has to record where and when the task ran, in which cooling zone, and through what thermal response its heat was removed.

Storage and demand-side flexibility complicate the picture further. Data centers are increasingly framed as integrated energy hubs with supply, storage, waste-heat, and demand-response capability [26], and as flexibility resources valuable to high-renewable power systems [27]—but that flexibility is not inherently low-carbon, since its emission effect depends on the marginal generation it displaces [28]. When storage participates in carbon-aware multi-energy coordination, it shifts energy across time and interacts with carbon-market decisions [29].

Shared electric-hydrogen storage and coupled carbon–green-certificate trading further introduce allocation decisions among multiple data-center prosumers [30], while different flexibility resources contribute unevenly to emissions, PUE, computing efficiency, and cost [31]. Among the studies reviewed above, none explicitly maintains a task-claimable low-carbon attribute inventory inside the storage device. The model developed here therefore records whether a nighttime discharge is backed by low-carbon electricity charged earlier and prevents the same attribute from being claimed more than once.

Meanwhile, low-carbon computing services are migrating from dispatch models toward platform practice. Reviews of the sociotechnical dynamics among AI, data centers, and power systems [32] and of sustainable data-center operation [33] describe the governance and engineering context; carbon-emission-aware scheduling has been implemented on Kubernetes [34], a spatio-temporal shifting algorithm has been proposed for edge-cloud environments [35], and real-time carbon-intensity data have been used to schedule heterogeneous servers [36]; distributed carbon-neutral operation [37] and DVFS-based DAG scheduling [38] show that task-graph structure and energy control can be represented in optimization models. These studies establish much of the engineering substrate for task-level carbon management. However, the studies reviewed above do not jointly formulate AI task execution trajectories, IT and cooling energy metering, a low-carbon attribute inventory in storage, unique low-carbon electricity claims, and task-level label risk within one dispatch model.

This gap defines the problem of the present paper. Decarbonization dispatch of an AI computing center park should answer not only how much total park emissions fall, but when a specific task is executed, how much traceable energy consumption it forms, how much low-carbon electricity claim it obtains, and which carbon-intensity label it earns. To this end, this paper proposes a decarbonization dispatch optimization method for AI computing center parks with task-level carbon traceability, in which task execution, energy-system operation, the low-carbon inventory in storage, and label risk are modeled in a unified manner. The main contributions are as follows.

1. A park decarbonization dispatch framework with consistent task, energy-consumption, and low-carbon ledgers is proposed. The framework handles AI task execution trajectories, resource-pool startup and shutdown, IT power consumption, network-storage energy consumption, cooling energy consumption, storage operation, and low-carbon electricity claims within the same dispatch boundary.
2. A task-level carbon traceability model is established. The task-level allocation relationships of dynamic IT energy consumption, resource-pool base energy consumption, thermally inertial cooling energy consumption, network-storage energy consumption, and storage losses are constructed, and duplicate allocation of green-electricity attributes is avoided through the unique low-carbon electricity claim constraint.
3. A stochastic dispatch model considering the tail violation risk of task-level low-carbon labels is constructed. The label-violation CVaR constraint is used to control the risk of label failure under adverse scenarios such as low renewable output and increased residual carbon factors.

The rest of this paper is organized as follows. Section 2 presents the task-energy coupled dispatch framework. Section 3 establishes the task-level carbon traceability model. Section 4 constructs the decarbonization dispatch optimization model. Section 5 conducts the case studies. Section 6 concludes the paper.

2. Task-energy coupled dispatch framework for AI computing center parks

Decarbonization dispatch of AI computing center parks must represent constraints that are absent from routine park energy optimization. These include job arrivals, dependencies, deadlines, heterogeneity, and service-quality requirements [8–10], as well as nonlinear cross-system energy characteristics and zonal cooling dynamics [19,22]. Consequently, AI task execution trajectories endogenously affect server IT load, cooling load, network-storage energy consumption, energy-storage operation, and grid-purchase demand. In the framework proposed here, the low-carbon claim of an individual AI task is jointly verified by task logs, energy-consumption metering, and low-carbon electricity sources.

Accordingly, this paper characterizes park dispatch as a consistency optimization problem among three ledgers: task execution, physical energy flow, and low-carbon claims. The task-execution ledger records the execution trajectories of tasks across periods and computing resource pools. The physical-energy-flow ledger records energy flows among servers, cooling, network storage, energy storage, auxiliary systems, and the grid. The low-carbon-claim ledger records the unique allocation relationship of claimable low-carbon electricity among specific tasks. Under this framework, low-carbon computing is no longer extrapolated from the total green-electricity share of the park, but is formed as a task-level traceable attribute through dispatch decisions and engineering metering. The overall task–energy coupled dispatch framework is illustrated in Fig. 1.

2.1. Park system structure and traceable energy flow

The AI computing center park considered in this paper includes AI server clusters, a cooling system, an energy storage system, park renewable energy, grid electricity purchase, and auxiliary energy-consuming units. The dispatch horizon, state periods, computing resource pools, cooling zones, and resource-pool affiliation relationships all follow the indexing system listed in the nomenclature.

In engineering implementation, the park can obtain the main operating data through grid-connection meters, renewable inverter meters, energy-storage PCS meters, rack PDUs, server telemetry data, network-storage meters, and cooling-branch meters. The same park metering boundary is adopted for carbon accounting. It covers the operational electricity consumption of IT equipment, cooling systems, network and storage equipment, auxiliary facilities, and energy-storage losses, while embodied emissions from servers, buildings, cooling equipment, and energy-storage manufacturing are outside the present scope. Task-level emissions represent the portions of facility-scope electricity emissions assigned to individual tasks according to their traceable energy consumption. Electricity use that cannot be assigned to a specific task remains in the background ledger; therefore, task-level emissions and background emissions together constitute the park-level total.

Throughout this paper, P-type variables denote period-average electrical power in MW, Q^{heat}-type variables denote thermal power in MW, and E-type variables and S-type state variables denote electrical energy in MWh. The duration of one dispatch period is denoted by Δt, in hours. Accordingly, $P_t\Delta t$ converts period-average power into period energy, whereas $Q^{heat}_{z,t,\Delta t}/\mathrm{COP}_{z,t,s}$ gives cooling electricity consumption. Low-carbon electricity inflow, storage inventory, and task-level low-carbon claims are also measured in MWh. The subscripts j, g, z, t, and s respectively identify an AI task, computing resource pool, cooling zone, dispatch period, and operating scenario. A superscript describes the physical or accounting role of a variable, such as grid purchase, renewable use, charging, discharging, cooling, or low-carbon attribution.

$$P_t^{\text{grid}} + P_t^{\text{ren,use}} + P_t^{\text{dis}} = P_t^{\text{IT}} + P_t^{\text{cool}} + P_t^{\text{ch}} + P_t^{\text{aux}} \tag{1}$$

Eq. (1) gives the power-conservation relationship within the park metering boundary. In (1), P_t^{grid}, $P_t^{\text{ren,use}}$, and P_t^{dis} denote grid-purchased, locally consumed renewable, and storage-discharging power, respectively. P_t^{IT}, P_t^{cool}, P_t^{ch}, and P_t^{aux} denote IT, cooling, storage-charging, and

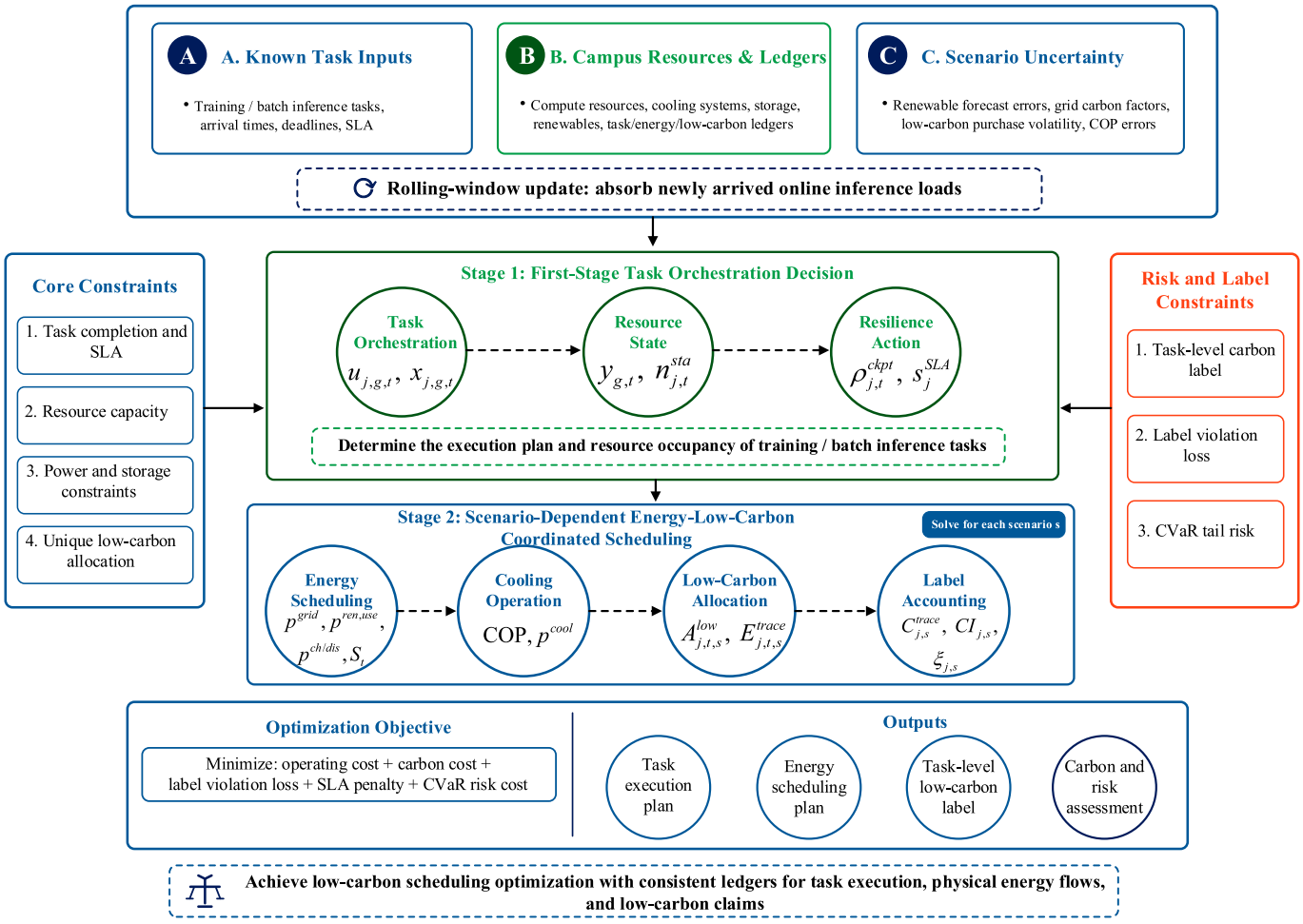


Fig. 1. Task–energy coupled decarbonization dispatch framework for AI computing center parks.

auxiliary power. All terms in (1) are period-average electrical power in MW.

The auxiliary power here is not a fixed exogenous constant, but consists of base auxiliary power and task-related network-storage power:

$$P_t^{\mathrm{aux}} = P_t^{\mathrm{aux,base}} + \frac{\sum_{j\in J} E_{j,t}^{\mathrm{net}} + E_{0,t}^{\mathrm{net}}}{\Delta t} \tag{2}$$

Here, $P^{\mathrm{aux,base}}_t$ is the background auxiliary power, $E_{j,t}^{\mathrm{net}}$ is the network-storage energy associated with task j, $E_{0,t}^{\mathrm{net}}$ is the corresponding background energy, J is the task set, and Δt is the dispatch-period duration in hours.

In (2), $E_{0,t}^{\mathrm{net}}$ is the measured or forecast energy parameter of background network, storage, and auxiliary systems, and is not freely selected by the optimizer. This treatment avoids physical non-closure caused by including task-related network-storage energy consumption in the task energy-consumption ledger while still treating P_t^{aux} as a fixed exogenous term in the park power balance.

This paper models physical electricity and claimable low-carbon electricity separately. The electricity entering the park low-carbon ledger consists of the actually consumed local renewable electricity and time-matched externally purchased low-carbon electricity:

$$E_t^{\mathrm{low,in}} = P_t^{\mathrm{ren,use}}\Delta t + E_t^{\mathrm{low,grid}} \tag{3}$$

If externally purchased low-carbon electricity is used as a dispatch decision, it must satisfy

$$0 \le E_t^{\mathrm{low,grid}} \le \bar{E}_t^{\mathrm{low,grid}}, \qquad E_t^{\mathrm{low,grid}} \le P_t^{\mathrm{grid}}\Delta t \tag{4}$$

Eq. (4) ensures that externally purchased low-carbon electricity does not exceed the contract or market procurement upper limit, nor the physical grid electricity purchased in that period, thereby preventing externally purchased low-carbon electricity from becoming an unlimited and cost-free virtual green-electricity source.

After considering the charging and releasing of low-carbon attributes through energy storage, the low-carbon claimable electricity that can be allocated to tasks in period t is

$$E_t^{\mathrm{low,avl}} = E_t^{\mathrm{low,in}} - E_t^{\mathrm{low,ch}} + E_t^{\mathrm{low,dis}} \tag{5}$$

The recursive relationship of the low-carbon energy inventory in storage is

$$S_{t+1}^{\mathrm{low}} = S_t^{\mathrm{low}} + n^{\mathrm{ch}}E_t^{\mathrm{low,ch}} - \frac{E_t^{\mathrm{low,dis}}}{n^{\mathrm{dis}}} \tag{6}$$

$$0 \le E_t^{\mathrm{low,ch}} \le E_t^{\mathrm{low,in}}, \qquad 0 \le E_t^{\mathrm{low,ch}} \le P_t^{\mathrm{ch}}\Delta t \tag{7}$$

$$0 \le E_t^{\mathrm{low,dis}} \le P_t^{\mathrm{dis}}\Delta t, \qquad 0 \le E_t^{\mathrm{low,dis}} \le n^{\mathrm{dis}}S_t^{\mathrm{low}}, \qquad 0 \le S_t^{\mathrm{low}} \le S_t \tag{8}$$

In (5)–(8), $E_t^{\mathrm{low,in}}$, $E_t^{\mathrm{low,ch}}$, $E^{\mathrm{low,dis}}_t$, and $E_t^{\mathrm{low,avl}}$ denote the low-carbon electricity entering the ledger, attributed to storage charging, released with storage discharging, and available for allocation, respectively. S_t^{low} is the remaining low-carbon attribute inventory, and S_t is the physical stored-energy state. n^{ch} and n^{dis} are the charging and discharging efficiencies, respectively.

Eqs. (6)–(8) restrict the source of low-carbon attributes in storage: storage can only release low-carbon electricity that has been actually charged and has not yet been released, and it cannot amplify low-carbon claims independently of the physical inventory.

The IT power of AI servers consists of resource-pool base power consumption and task-execution power consumption:

$$P_t^{\mathrm{IT}} = \sum_{g\in G}\left(P_g^{\mathrm{idle}} y_{g,t} + \sum_{j\in J} k_{j,g}^{\mathrm{it}} u_{j,g,t}\right) \tag{9}$$

The computing-capacity constraint of resource pools is

$$\sum_{j\in J} u_{j,g,t} \le R_g y_{g,t}, \qquad g \in G,\ \ t \in T \tag{10}$$

Here, $u_{j,g,t}$ is the workload executed by task j in resource pool g during period t, $x_{j,g,t}$ is the corresponding resource-pool assignment indicator, $y_{g,t}$ is the operating-state indicator of resource pool g, R_g is its available computing-service capacity, and $b_{j,t}$ indicates whether task j is running.

Eqs. (9) and (10) show that the load of an AI computing center park is not an exogenous curve, but is endogenously formed by task execution, resource-pool startup and shutdown, and computing allocation.

2.2. AI task types and traceable dispatch characteristics

Data-center workloads can be distinguished by latency tolerance and execution constraints. Existing scheduling and load-modeling studies distinguish delay-sensitive or interactive workloads from periodic or latency-tolerant batch jobs and explicitly model arrival patterns, dependencies, quality-of-service requirements, and temporal shifting [8,9,19]. In this paper, these workload classes are instantiated as training tasks, batch inference tasks, and online inference tasks. Training tasks are modeled with relatively large workloads and wider completion windows and may participate in rolling decarbonization dispatch through delayed startup, segmented execution, or checkpoint mechanisms. Batch inference tasks are modeled as delay-tolerant but subject to batch-completion deadlines. Online inference tasks are treated as latency-sensitive and non-transferable across periods; they are aggregated into rigid online-inference baseline loads or base auxiliary service loads. The optimization model mainly characterizes training and batch inference tasks that have entered the dispatch window and have explicit arrival times and deadlines. Stochastic online arrival requests are converted into new known tasks or rigid baseline loads when the rolling window is updated.

For task j, a_j and d_j denote its arrival and deadline periods, and N_T denotes the number of periods in the dispatch horizon; hence, $1 \le a_j \le d_j \le N_T$. The task execution and resource-occupation relationships are as follows:

$$0 \le u_{j,g,t} \le R_g x_{j,g,t}, \qquad x_{j,g,t} \le y_{g,t}, \qquad b_{j,t} = \sum_{g\in G} x_{j,g,t} \tag{11}$$

$$\sum_{g\in G} x_{j,g,t} \le 1, \qquad j \in J,\ \ t \in T \tag{12}$$

This paper assumes that a single task can occupy at most one resource pool in a single period. If parallel training tasks are studied, $b_{j,t}$ can be rewritten as a running indicator variable, and (12) can be extended to a multi-resource-pool parallel occupation constraint.

To avoid misclassifying the first startup or normal shutdown as checkpoint recovery loss, this paper explicitly distinguishes startup, shutdown, restart after interruption, and cross-resource-pool migration. Taking $b_{j,0} = 0$ and $x_{j,g,0} = 0$, as follows:

$$n_{j,t}^{\mathrm{sta}} \ge b_{j,t} - b_{j,t-1}, \quad n_{j,t}^{\mathrm{sta}} \le b_{j,t}, \quad n_{j,t}^{\mathrm{sta}} \le 1 - b_{j,t-1} \tag{13}$$

$$n_{j,t}^{\mathrm{stop}} \ge b_{j,t-1} - b_{j,t}, \quad n_{j,t}^{\mathrm{stop}} \le b_{j,t-1}, \quad n_{j,t}^{\mathrm{stop}} \le 1 - b_{j,t} \tag{14}$$

A shutdown event is used only to record changes in operating status or to extend constraints such as minimum running time, and does not enter the checkpoint loss term. The historical startup indicator is given recursively to avoid unnecessary constraint scale caused by summing all historical periods one by one:

$$h_{j,a_j}^{\mathrm{hist}} = 0, \qquad h_{j,t+1}^{\mathrm{hist}} \ge h_{j,t}^{\mathrm{hist}}, \qquad h_{j,t+1}^{\mathrm{hist}} \ge n_{j,t}^{\mathrm{sta}}, \qquad t = a_j, \ldots, d_j \tag{15}$$

$$h_{j,t+1}^{\mathrm{hist}} \le h_{j,t}^{\mathrm{hist}} + n_{j,t}^{\mathrm{sta}}, \qquad t = a_j, \ldots, d_j \tag{16}$$

Eqs. (15)–(16) reduce the number of historical startup-state constraints from $O(N2\,T)$ to $O(N_T)$. The restart-after-interruption variable is jointly triggered by the startup variable and the historical startup state:

$$q_{j,t}^{\mathrm{rst}} \le n_{j,t}^{\mathrm{sta}}, \quad q_{j,t}^{\mathrm{rst}} \le h_{j,t}^{\mathrm{hist}}, \quad q_{j,t}^{\mathrm{rst}} \ge n_{j,t}^{\mathrm{sta}} + h_{j,t}^{\mathrm{hist}} - 1 \tag{17}$$

Cross-resource-pool migration is exactly linearized through the transition variable:

$$q_{j,g,g',t}^{\mathrm{tr}} \le x_{j,g,t-1}, \quad q_{j,g,g',t}^{\mathrm{tr}} \le x_{j,g',t}, \quad q_{j,g,g',t}^{\mathrm{tr}} \ge x_{j,g,t-1} + x_{j,g',t} - 1, \quad g \ne g' \tag{18}$$

$$q_{j,t}^{\mathrm{mig}} = \sum_{g\in G}\sum_{g'\in G: g'\ne g} q_{j,g,g',t}^{\mathrm{tr}}, \qquad t > a_j \tag{19}$$

Eqs. (18)–(19) provide the upper and lower bounds of $q_{j,t}^{\mathrm{mig}}$, preventing it from being driven indirectly to zero only by the objective function when $L_j^{\mathrm{ckpt}} > 0$ or $c_j^{\mathrm{ckpt}} > 0$. Finally, checkpoint loss is triggered only by restart after interruption and cross-resource-pool migration:

$$q_{j,t}^{\mathrm{ckpt}} \ge q_{j,t}^{\mathrm{rst}}, \qquad q_{j,t}^{\mathrm{ckpt}} \ge q_{j,t}^{\mathrm{mig}}, \qquad q_{j,t}^{\mathrm{ckpt}} \le q_{j,t}^{\mathrm{rst}} + q_{j,t}^{\mathrm{mig}} \tag{20}$$

Let W_j denote the required computational workload of task j. The task progress is updated only within its executable window:

$$r_{j,t+1} = r_{j,t} + \sum_{g\in G} u_{j,g,t} d_t - L_j^{\mathrm{ckpt}} q_{j,t}^{\mathrm{ckpt}}, \qquad t = a_j, \ldots, d_j \tag{21}$$

$$L_j^{\mathrm{ckpt}} q_{j,t}^{\mathrm{ckpt}} \le r_{j,t}, \qquad t = a_j, \ldots, d_j \tag{22}$$

$$0 \le r_{j,t} \le W_j, \qquad t \in T^{+} \tag{23}$$

$$r_{j,t} = 0 \quad (t \le a_j), \qquad r_{j,d_j+1} + s_j^{\mathrm{SLA}} \ge W_j, \qquad s_j^{\mathrm{SLA}} \ge 0 \tag{24}$$

For tasks applying for low-carbon labels, this paper requires that labels be issued only for completed tasks, namely

$$s_j^{\mathrm{SLA}} = 0, \qquad j \in J^{\mathrm{low}} \tag{25}$$

Tasks can run only within their allowed windows:

$$u_{j,g,t} = 0, \quad x_{j,g,t} = 0, \quad b_{j,t} = 0, \qquad t < a_j \ \text{ or } \ t > d_j \tag{26}$$

If task j does not allow interruption or migration, $\sum_{t\in T} n_{j,t}^{\mathrm{sta}} \le 1$ and $q_{j,t}^{mig} = 0$ can be added. If checkpoint is allowed, (21) also reflects the progress loss caused by restart or migration.

This paper adopts a unified dimensional convention: equivalent computing-service rate, computational workload, and checkpoint loss use the same workload basis; energy, power, and emission factors maintain consistent unit conversions. This ensures dimensional consistency among the progress equation, power balance, and carbon-intensity calculation.

Task-level carbon traceability also needs to allocate resource-pool base power consumption and dynamic power consumption to specific tasks. The base-power allocation coefficient is predetermined by the audit rule, cannot be arbitrarily manipulated by the optimizer, and cannot allocate energy consumption and green electricity to unexecuted tasks. Therefore, this paper adopts a traceability protocol of "fixed rule with execution support taking effect", and it satisfies

$$0 \le f^{\text{idle}}_{j,g,t} \le 1, \qquad \sum_{j\in J} f^{\text{idle}}_{j,g,t} \le 1, \qquad g \in G,\ \ t \in T \tag{27}$$

The IT energy of task j in period t is expressed as

$$E^{\text{IT}}_{j,t} = d_t \sum_{g\in G} \left(k^{\text{it}}_{j,g} u_{j,g,t} + f^{\text{idle}}_{j,g,t} P^{\text{idle}}_{g} x_{j,g,t} \right). \tag{28}$$

Resource-pool base energy consumption that is not occupied by specific AI tasks is included in the background or common energy-consumption ledger $E^{trace}_{0,t}$. Therefore, the dispatchable boundary of task j can be written as the projection of this task onto the global feasible region:

$$O^{\text{sch}}_{j} = \text{proj}_{j}\{(u,x,b,n,h,q,r,s):\ \ (11)-(26),(10),(27)\} \tag{29}$$

The larger the projected feasible region is, the more easily the task can be shifted to periods with sufficient low-carbon electricity, higher cooling efficiency, or lower grid carbon factors. The smaller the projected feasible region is, the closer the task is to a rigid load, and the more its label result depends on the energy-supply capability of the park.

2.3. Traceable energy mapping for AI tasks

In this paper, task-level carbon traceability is defined as a consistency mapping among task execution trajectories, metered energy consumption, and low-carbon electricity claims rather than as a post-event proportional allocation of total park emissions. This mapping imposes three engineering requirements: the heat generated by task execution is converted into cooling electricity consumption; cooling energy is zonal and thermally inertial and is not allocated using only the average park PUE; and low-carbon electricity is treated as a finite claimable resource that must be matched with actual task execution without duplicate claims.

For the heat load generated by task j in cooling zone z, define

$$Q^{\text{heat}}_{j,z,t} = \sum_{g:z(g)=z} k^{\text{heat}}_{g} \left(k^{\text{it}}_{j,g} u_{j,g,t} + f^{\text{idle}}_{j,g,t} P^{\text{idle}}_{g} x_{j,g,t} \right) \tag{30}$$

Let k denote the rolling-window index and τ_k denote the first global period of window k. The thermal response inherited from preceding windows is retained at the task and cooling-zone levels. For task j, the historical thermal-response term entering period t of window k is

$$Q^{\text{hist},k}_{j,z,t} = \sum_{l=t}^{H_z} m^{\text{th}}_{z,l} \widehat{Q}^{\text{heat}}_{j,z,\tau_k+t-l-1}, \qquad t \le H_z \tag{31}$$

and $Q^{\text{hist},k}_{j,z,t}=0$ for $t > H_z$. Here, $\widehat{Q}^{\text{heat}}_{j,z,\tau}$ denotes the realized task heat load recorded before the current window. The cooling electricity allocated to task j in period t is therefore

$$E^{\text{cool},k}_{j,t,s} = \Delta t \sum_{z\in Z} \frac{Q^{\text{hist},k}_{j,z,t} + \sum_{l=0}^{\min\{H_z,t-1\}} m^{\text{th}}_{z,l} Q^{\text{heat},k}_{j,z,t-l}}{\text{COP}_{z,t,s}} \tag{32}$$

The first term represents cooling electricity inherited from heat generated before the current window, whereas the second term represents the response to heat generated within the current window. After each implemented period, the realized task heat load is added to the thermal-history record and used to construct $Q^{\text{hist},k+1}_{j,z,t}$ for the next rolling window. Hence, the cooling load generated by a task near the end of one window is retained under the same task identity and settled in the subsequent window rather than being reassigned to the background ledger.

Furthermore, (32) can be summarized as a matrix form of the thermal-inertia convolution mapping. For any cooling zone z, define the heat-load time vector of task j as

$$\mathbf{q}^{\text{heat}}_{j,z} = \left[Q^{\text{heat}}_{j,z,1}, Q^{\text{heat}}_{j,z,2}, \ldots, Q^{\text{heat}}_{j,z,N_T} \right]^{\top}, \qquad \mathbf{q}^{\text{task}}_{z} = \sum_{j\in J} \mathbf{q}^{\text{heat}}_{j,z} \tag{33}$$

A causal lower-triangular Toeplitz thermal response matrix $\mathbf{H}^{\text{th}}_{z} \in \mathbb{R}^{N_T\times N_T}$ is constructed from the thermal response kernel $m^{\text{th}}_{z,l}$:

$$\left[\mathbf{H}^{\text{th}}_{z}\right]_{t,r} = \begin{cases} m^{\text{th}}_{z,t-r}, & 0 \le t-r \le H_z, \\ 0, & \text{otherwise}, \end{cases} \qquad t,r \in T \tag{34}$$

namely

$$H^{\text{th}}_{z} = \begin{bmatrix} m^{\text{th}}_{z,0} & 0 & 0 & \cdots & 0 \\ m^{\text{th}}_{z,1} & m^{\text{th}}_{z,0} & 0 & \cdots & 0 \\ \vdots & \vdots & \ddots & \ddots & \vdots \\ m^{\text{th}}_{z,H_z} & m^{\text{th}}_{z,H_z-1} & \cdots & m^{\text{th}}_{z,0} & 0 \\ 0 & m^{\text{th}}_{z,H_z} & \cdots & m^{\text{th}}_{z,1} & m^{\text{th}}_{z,0} \\ \vdots & \vdots & \ddots & \vdots & \vdots \end{bmatrix} \tag{35}$$

Here, the matrix shows only the first several rows and typical middle rows. Elements outside the lag range 0,1, …,H_z are all set to zero, and the strict definition follows (34). Let

$$\mathbf{D}^{\text{COP}}_{z} = \text{diag}\left(\frac{d_1}{\text{COP}_{z,1}}, \frac{d_2}{\text{COP}_{z,2}}, \ldots, \frac{d_{N_T}}{\text{COP}_{z,N_T}} \right) \tag{36}$$

Then the incremental cooling-energy vector borne by task j in cooling zone z can be written as

$$\mathbf{e}^{\text{cool},k}_{j,z} = \mathbf{D}^{\text{COP},k}_{z} \left(\mathbf{H}^{\text{th}}_{z} \mathbf{q}^{\text{heat},k}_{j,z} + \mathbf{q}^{\text{hist},k}_{j,z} \right), \qquad \mathbf{e}^{\text{cool},k}_{j} = \sum_{z\in Z} \mathbf{e}^{\text{cool},k}_{j,z} \tag{37}$$

The task-level cooling-energy mapping is a positive affine mapping of the current-window task heat load, with the historical thermal-response vector treated as a fixed carry-over state. The lower-triangular Toeplitz matrix preserves causality within the current window, while the historical vector preserves the thermal response generated before the window. Both components retain the identity of the task that generated the heat load.

When $m^{\text{th}}_{z,l} \ge 0$, $\boldsymbol{H}^{\text{th}}_{z}$ is a nonnegative lower-triangular matrix, indicating that the current cooling responsibility is affected only by current and historical heat loads and is not affected by future task execution states. Further let

$$M^{\text{th}}_{z} = \sum_{l=0}^{H_z} m^{\text{th}}_{z,l}, \qquad 0 \le M^{\text{th}}_{z} \le 1 \tag{38}$$

Then the thermal response matrix has a monotone mapping property. For any two task heat-load vectors $\mathbf{q}1\ z \le \boldsymbol{q}^{2}_{z}$, we have

$$\mathbf{H}^{\text{th}}_{z} \mathbf{q}^{1}_{z} \le \mathbf{H}^{\text{th}}_{z} \mathbf{q}^{2}_{z} \tag{39}$$

That is, when the task heat load increases, its cooling responsibility will not decrease in the opposite direction. Meanwhile, for any $\mathbf{q}_z$, this mapping satisfies

$$\left\| \mathbf{H}^{\text{th}}_{z} \mathbf{q}_{z} \right\|_{1} \le M^{\text{th}}_{z} \|\mathbf{q}_{z}\|_{1}, \qquad \left\| \mathbf{H}^{\text{th}}_{z} \mathbf{q}_{z} \right\|_{\infty} \le M^{\text{th}}_{z} \|\mathbf{q}_{z}\|_{\infty} \tag{40}$$

When the finite-horizon thermal response kernel satisfies the normalization boundary, thermal-inertia traceability does not amplify task heat responsibility and does not generate nonphysical negative allocations. These properties are used to ensure the physical consistency and accounting stability of task-level cooling-energy traceability without adding new optimization decision variables.

Task-related network, storage, and checkpoint read/write energy consumption is expressed as

$$E^{\text{net}}_{j,t} = a^{\text{io}}_{j} D^{\text{io}}_{j,t} + a^{\text{sync}}_{j} D^{\text{sync}}_{j,t} + a^{\text{ckpt}}_{j} q^{\text{ckpt}}_{j,t} \tag{41}$$

where the data read/write volume and parameter synchronization volume are determined by task execution amount and startup/shutdown actions, and are not variables that can be freely compressed by the optimizer:

$$D_{j,t}^{\mathrm{io}} = k_j^{\mathrm{io}} \sum_{g\in G} u_{j,g,t} d_t + D_j^{\mathrm{load}} n_{j,t}^{\mathrm{sta}} + D_j^{\mathrm{save}} n_{j,t}^{\mathrm{stop}} + D_j^{\mathrm{save}} \mathbb{I}_{\{t=N_T\}} b_{j,N_T} \tag{42}$$

$$D_{j,t}^{\mathrm{sync}} = k_j^{\mathrm{sync}} \sum_{g\in G} u_{j,g,t} d_t \tag{43}$$

$$0 \le D_{j,t}^{\mathrm{io}} \le \overline{D}_j^{\mathrm{io}}, \qquad 0 \le D_{j,t}^{\mathrm{sync}} \le \overline{D}_j^{\mathrm{sync}}, \qquad j \in J,\ \ t \in T \tag{44}$$

Eqs. (42)–(43) ensure that execution-related network-storage energy consumption is generated with actual task operation and startup/shutdown actions.

Combining IT energy consumption, cooling energy consumption, network-storage energy consumption, and allocated storage losses, the traceable electricity of task j in period t is defined as

$$E_{j,t}^{\mathrm{trace}} = E_{j,t}^{\mathrm{IT}} + E_{j,t}^{\mathrm{cool}} + E_{j,t}^{\mathrm{net}} + E_{j,t}^{\mathrm{sto}} \tag{45}$$

To avoid arbitrary allocation of storage losses to a certain type of task, $E_{j,t}^{\mathrm{sto}}$ adopts a fixed and auditable allocation rule that is consistent with execution support. Let $\overline{b}_{j,t}^{\mathrm{sto}}$ be the fixed storage-loss allocation coefficient predetermined by the audit rule, and the parameter itself satisfies

$$0 \le \overline{b}_{j,t}^{\mathrm{sto}} \le 1, \qquad \sum_{j\in J} \overline{b}_{j,t}^{\mathrm{sto}} \le 1, \qquad t \in T \tag{46}$$

Let Msto t be a sufficiently large loss upper bound. Then the actual allocated energy $E_{j,t}^{\mathrm{sto}}$ satisfies

$$0 \le E_{j,t}^{\mathrm{sto}} \le \overline{b}_{j,t}^{\mathrm{sto}} E_t^{\mathrm{sto,loss}}, \qquad E_{j,t}^{\mathrm{sto}} \le M_t^{\mathrm{sto}} b_{j,t} \tag{47}$$

$$E_{j,t}^{\mathrm{sto}} \ge \overline{b}_{j,t}^{\mathrm{sto}} E_t^{\mathrm{sto,loss}} - M_t^{\mathrm{sto}} \left(1 - b_{j,t}\right), \qquad \sum_{j\in J} E_{j,t}^{\mathrm{sto}} \le E_t^{\mathrm{sto,loss}} \tag{48}$$

Eqs. (46)–(48) ensure that tasks that are not running do not bear storage losses, running tasks are allocated according to fixed coefficients, and the sum of allocation coefficients over all tasks in any period does not exceed 1. The unallocated part enters the background residual ledger, thereby avoiding confusion between the identities of "allocation parameters" and "dispatch decisions". The equivalent storage loss energy is

$$E_t^{\mathrm{sto,loss}} = \left(1 - \eta^{\mathrm{ch}}\right) P_t^{\mathrm{ch}} d_t + \left(\frac{1}{\eta^{\mathrm{dis}}} - 1\right) P_t^{\mathrm{dis}} d_t \tag{49}$$

The total allocable energy within the park metering boundary is defined as

$$E_t^{\mathrm{meter}} = \left(P_t^{\mathrm{IT}} + P_t^{\mathrm{cool}} + P_t^{\mathrm{aux}}\right) d_t + E_t^{\mathrm{sto,loss}} \tag{50}$$

and it is required that

$$\sum_{j\in J} E_{j,t}^{\mathrm{trace}} + E_{0,t}^{\mathrm{trace}} = E_t^{\mathrm{meter}}, \qquad E_{0,t}^{\mathrm{trace}} \ge 0 \tag{51}$$

Eq. (50) clarifies the physical meaning of E_t^{meter}: storage charging and discharging itself is energy time shifting and is not directly counted again as task energy use; only storage losses enter the allocable electricity, thereby avoiding double counting in the energy ledger.

Low-carbon claims must satisfy execution consistency and unique allocation relationships:

$$0 \le A_{j,t}^{\mathrm{low}} \le E_{j,t}^{\mathrm{trace}}, \qquad 0 \le A_{0,t}^{\mathrm{low}} \le E_{0,t}^{\mathrm{trace}} \tag{52}$$

$$\sum_{j\in J} A_{j,t}^{\mathrm{low}} + A_{0,t}^{\mathrm{low}} \le E_t^{\mathrm{low,avl}} \tag{53}$$

Eq. (52) means that a task is eligible to obtain a low-carbon claim only when it actually forms traceable energy consumption in the corresponding period. Eq. (53) means that the claimable low-carbon electricity in the same period can be allocated only once and cannot be reused by multiple tasks.

To handle the residual mixed electricity after local renewable energy, externally purchased low-carbon electricity, and ordinary purchased electricity jointly supply energy, the traceable carbon emissions of task j are

$$C_j^{\mathrm{trace}} = \sum_{t\in T} \left[EF_{t,k}^{\mathrm{low}} A_{j,t}^{\mathrm{low}} + EF_t^{\mathrm{res}} \left(E_{j,t}^{\mathrm{trace}} - A_{j,t}^{\mathrm{low}} \right) \right] \tag{54}$$

When residual mixed electricity is not distinguished, EFres t can be set equal to the carbon factor of ordinary grid-purchased electricity. If multiple low-carbon sources exist simultaneously and have different carbon factors, Alow j,t can be further split by source into $A_{j,t,k}^{\mathrm{low}}$ and the corresponding $EF_{t,k}^{\mathrm{low}}$ can be used. This residual-mix-factor treatment avoids systematic overestimation or underestimation of the remaining electricity that has not been claimed or cannot be accurately matched. $E_{j,s}^{\mathrm{trace}}$ is an exogenous standard parameter for market-based carbon accounting and is not inferred by the dispatch model.

Task-level carbon intensity is defined as

$$CI_j = \frac{C_j^{\mathrm{trace}}}{W_j} \tag{55}$$

Eq. (25) ensures that CI_j is calculated and issued only after tasks applying for low-carbon labels are completed. In summary, task-level carbon traceability for AI computing center parks can be summarized as the following three-ledger consistency relationship:

$$\begin{cases} \text{Task ledger}: r_{j,t+1} = r_{j,t} + \sum_{g\in G} u_{j,g,t} d_t - L_j^{\mathrm{ckpt}} q_{j,t}^{\mathrm{ckpt}}, \\ \text{Energy ledger}: \sum_{j\in J} E_{j,t}^{\mathrm{trace}} + E_{0,t}^{\mathrm{trace}} = E_t^{\mathrm{meter}}, \\ \text{Low-carbon ledger}: \sum_{j\in J} A_{j,t}^{\mathrm{low}} + A_{0,t}^{\mathrm{low}} \le E_t^{\mathrm{low,avl}}, 0 \le A_{j,t}^{\mathrm{low}} \le E_{j,t}^{\mathrm{trace}}. \end{cases} \tag{56}$$

Eq. (56) summarizes the ledger-closure relationship of the proposed framework: dispatch decisions change task execution time and energy-system operation, but task execution records, physical energy-consumption metering, and low-carbon electricity claims must remain consistent.

3. Task-level carbon traceability model

3.1. Traceability of IT, cooling, and auxiliary energy consumption

For task j, its dynamic IT energy on resource pool g in period t is

$$E_{j,g,t}^{\mathrm{dyn}} = k_{j,g}^{\mathrm{it}} u_{j,g,t} d_t \tag{57}$$

The base operating energy consumption of a resource pool is generated by servers, network interfaces, memory, and local storage that are in the on state. According to the fixed traceability protocol given in Section 2, the base IT energy allocated to task j is

$$E_{j,g,t}^{\mathrm{idle}} = f_{j,g,t}^{\mathrm{idle}} P_g^{\mathrm{idle}} x_{j,g,t} d_t \tag{58}$$

Therefore, the traceable IT energy of task j in period t is

$$E_{j,t}^{\mathrm{IT}} = \sum_{g\in G} \left(E_{j,g,t}^{\mathrm{dyn}} + E_{j,g,t}^{\mathrm{idle}} \right) \tag{59}$$

3.2. Storage losses and low-carbon attribute allocation

Storage losses under a scenario or an actual operating path are given by (49), and task-level allocation is given by (47)–(48). If it cannot be demonstrated that task j benefits from storage operation within a settlement period, $\bar{b}^{\mathrm{sto}}_{j,t}, t=0$ should be set, and the corresponding losses enter $E^{trace}_{0,t}$. This treatment prevents the optimizer from arbitrarily transferring the carbon-emission responsibility corresponding to storage losses.

The formation of task-level carbon labels depends on low-carbon electricity attributes. Low-carbon claims need to simultaneously satisfy execution consistency and unique-claim constraints, namely (52)–(53). Combining IT, cooling, storage, network, and storage energy consumption, the traceable energy of task j in period t is given by (45). If multiple low-carbon sources are adopted, a source index k can be further introduced, and $\sum_k A^{\mathrm{low}}_{j,t,k} \le E^{\mathrm{trace}}_{j,t}$ can be imposed to support more fine-grained carbon-factor accounting.

3.3. Task-level carbon labels and traceability consistency constraints

After task-level traceable electricity is obtained, the traceable carbon emissions and carbon intensity of task j are given by (54) and (55), respectively. To support grading of low-carbon computing services, task carbon intensity can be mapped into discrete labels:

$$\mathrm{Label}^{\mathrm{post}}_j = \begin{cases} \mathrm{A}, & CI_j \le B_1, \\ \mathrm{B}, & B_1 < CI_j \le B_2, \\ \mathrm{C}, & B_2 < CI_j \le B_3, \\ \mathrm{D}, & CI_j > B_3. \end{cases} \tag{60}$$

Eq. (60) is a post-processing rule for result interpretation and settlement. In the optimization model, the direct constraints are the upper limit of task carbon intensity $\overline{CI}_j$ and its violation slack variable. The final label can be generated based on the realized path, worst-case scenario, or expected carbon intensity.

To avoid task-level carbon labels degenerating into post-event allocation results, this paper imposes traceability consistency constraints. First, task-level energy traceability should be consistent with park-metered electricity:

$$\sum_{j\in J} E^{\mathrm{trace}}_{j,t} + E^{\mathrm{trace}}_{0,t} = E^{\mathrm{meter}}_t, \qquad t \in T \tag{61}$$

Second, task-level low-carbon claims should be consistent with claimable low-carbon electricity:

$$\sum_{j\in J} A^{\mathrm{low}}_{j,t} + A^{\mathrm{low}}_{0,t} \le E^{\mathrm{low,avl}}_t, \qquad t \in T \tag{62}$$

Finally, the aggregation of task carbon emissions should remain consistent with the park carbon-accounting boundary:

$$\sum_{j\in J} C^{\mathrm{trace}}_j + C^{\mathrm{trace}}_0 = C^{\mathrm{park}} \tag{63}$$

4. Decarbonization dispatch optimization model considering task-level carbon traceability

This section embeds task-level traceable electricity, unique low-carbon electricity claims, and task-level carbon labels into the dispatch model, so that the model can generate verifiable task-level low-carbon labels while satisfying power-balance and task-service constraints.

Considering forecast errors in renewable generation, grid carbon factors, externally purchased low-carbon electricity, and cooling efficiency, scenario probabilities are assumed to be nonnegative and to sum to one. This paper adopts a rolling two-stage risk-aware dispatch structure, consistent with scenario-based recourse and two-stage data-center energy-management models [21,30]. Within each rolling window, training and batch inference tasks that have entered the window constitute the known task set, and task-orchestration variables are first-stage decision variables. Grid electricity purchase, storage operation, low-carbon electricity claims, and traceability accounting variables are adjusted with scenario s. Stochastically arriving online inference loads are not directly modeled in this static task set, but are updated as newly arrived tasks or rigid baseline loads in the next rolling window. If task-arrival uncertainty needs to be represented in the same model, task variables can be extended to scenario-dependent second-stage recourse variables, and nonanticipativity constraints can be imposed on scenarios sharing the same history information. To highlight the coupling mechanism between task-level carbon traceability and energy dispatch, this paper adopts the above rolling two-stage form $\sum_{s\in S} p_s = 1$.

Each rolling optimization uses a 24-h prediction window and a 1-h rolling step. Only the first-period decisions are implemented and written into the task, energy, and low-carbon ledgers; decisions in the remaining prediction periods are reoptimized in the next window and are not repeatedly recorded. The realized task heat load in the implemented period is transferred to the task-level thermal-history state of the next window. A task that has completed its computational workload remains in the tail-settlement ledger until its inherited thermal response becomes zero. During this interval, its execution variables are fixed to zero, while the subsequent cooling electricity, low-carbon electricity claims, and carbon emissions continue to be assigned to the same task.

The proposed risk-aware decarbonization dispatch optimization framework is illustrated in Fig. 2.

4.1. Objective function and dispatch decision variables

The dispatch objective is

$$\min F = \sum_{s\in S} p_s \left(F^{\mathrm{rec}}_s + c^{\mathrm{park}} C^{\mathrm{park}}_s + c^{\mathrm{tag}} V^{\mathrm{tag}}_s\right) + \sum_{j\in J}\sum_{t\in T} c^{\mathrm{ckpt}}_j q^{\mathrm{ckpt}}_{j,t} + c^{\mathrm{SLA}} \sum_{j\in J} w^{\mathrm{SLA}}_j s^{\mathrm{SLA}}_j + c^{\mathrm{risk}} R^{\mathrm{tag}} \tag{64}$$

The deterministic checkpoint cost and SLA shortfall penalty are placed outside the scenario summation. If they are written into the scenario expectation term, the numerical value is equivalent because $\sum_s p_s = 1$, but the model structure is less clear than (64). The coefficients in (64) were specified before optimization and kept unchanged across all comparison methods, operating scenarios, and test systems. The economic terms retain their original monetary units, while park carbon emissions, label violations, SLA shortfalls, and label tail risk are converted into monetary-equivalent costs through their respective coefficients. Therefore, no additional benchmark or min–max normalization is applied. This treatment preserves the relative economic scale of the different objectives and avoids changing the weighting coefficients with the test-system size. The adopted coefficients and their units are reported in Table 1.

The recourse operating cost under scenario s is

$$\begin{aligned} F^{\mathrm{rec}}_s = & \sum_{t\in T} c^{\mathrm{grid}}_{t,s} P^{\mathrm{grid}}_{t,s} d_t + \sum_{t\in T} c^{\mathrm{low,grid}}_{t,s} E^{\mathrm{low,grid}}_{t,s} \\ & + \sum_{t\in T} c^{\mathrm{sto}} \left(P^{\mathrm{ch}}_{t,s} + P^{\mathrm{dis}}_{t,s}\right) d_t + \sum_{t\in T} c^{\mathrm{cur}} P^{\mathrm{cur}}_{t,s} d_t \\ & + \sum_{z\in Z}\sum_{t\in T} c^{\mathrm{cool}} v^{\mathrm{cool}}_{z,t,s} + c^{\mathrm{low,ter}} v^{\mathrm{low,ter}}_s \end{aligned} \tag{65}$$

In (65), $c^{\mathrm{low,grid}}_{t,s}$ only represents the low-carbon attribute premium, REC cost, or PPA price difference. The physical electricity cost has already been included by $c^{\mathrm{grid}}_{t,s} P^{\mathrm{grid}}_{t,s} d_t$, and the two are not double-counted.

The total market-based emissions under scenario s are given as an accounting definition:

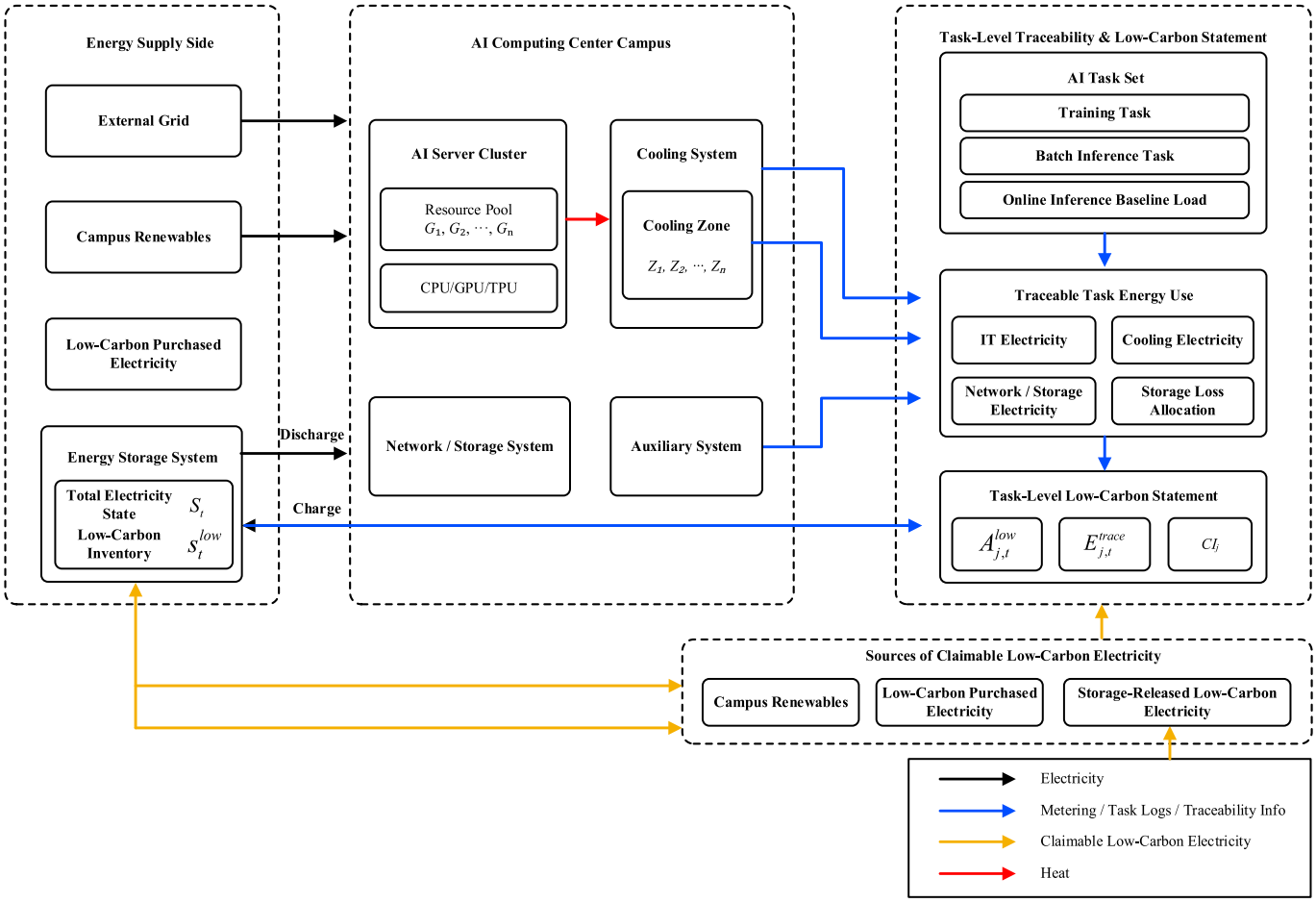


Fig. 2. Risk-aware decarbonization dispatch optimization framework with task-level carbon traceability.

$$C_s^{\text{park}} = \sum_{j\in J} C_{j,s}^{\text{trace}} + C_{0,s}^{\text{trace}} \tag{66}$$

This equation confines park-level carbon emissions and task-level traceable carbon emissions within the same accounting boundary, avoiding inconsistency between total market-based emissions and the aggregation of task labels.

4.2. Constraints of AI tasks, computing resources, and the park energy system

The task execution constraints follow (11)–(25). In Section 4, these constraints can be viewed as first-stage task-orchestration constraints. When a more refined multistage model is adopted, part of the startup or migration variables can also be set as second-stage variables, and non-anticipativity constraints can be used to maintain information-structure consistency. The capacity constraint of computing resource pools is

$$\sum_{j\in J} u_{j,g,t} \le R_g y_{g,t}, \qquad g \in G,\ \ t \in T \tag{67}$$

$$y_{g,t}, x_{j,g,t}, b_{j,t}, n_{j,t}^{\text{sta}}, n_{j,t}^{\text{stop}}, h_{j,t}^{\text{hist}}, q_{j,t}^{\text{rst}}, q_{j,t}^{\text{mig}}, q_{j,t}^{\text{ckpt}}, q_{j,g,g',t}^{\text{tr}} \in \{0,1\}, \quad u_{j,g,t} \ge 0, \quad r_{j,t} \ge 0, \quad s_j^{\text{SLA}} \ge 0 \tag{68}$$

The operation of the park power system under scenario s satisfies the power-balance constraint:

$$P_{t,s}^{\text{grid}} + P_{t,s}^{\text{ren,use}} + P_{t,s}^{\text{dis}} = P_t^{\text{IT}} + P_{t,s}^{\text{cool}} + P_{t,s}^{\text{ch}} + P_t^{\text{aux}}, \qquad t \in T,\ \ s \in S \tag{69}$$

The IT power and auxiliary power in (69) are given by (9) and (2), respectively. If network-storage energy consumption varies across scenarios, it can be extended to $P_{t,s}^{\text{aux}}$. Renewable-utilization power and grid-purchase power satisfy

$$0 \le P_{t,s}^{\text{ren,use}} \le \overline{P}_{t,s}^{\text{ren}}, \qquad P_{t,s}^{\text{cur}} = \overline{P}_{t,s}^{\text{ren}} - P_{t,s}^{\text{ren,use}} \tag{70}$$

$$0 \le P_{t,s}^{\text{grid}} \le \overline{P}^{\text{grid}} \tag{71}$$

Eqs. (70)–(71) jointly define the boundaries of renewable utilization, renewable curtailment, and grid electricity purchase.

The storage operation constraints are

$$S_{t+1,s} = S_{t,s} + n^{\text{ch}} P_{t,s}^{\text{ch}} d_t - \frac{P_{t,s}^{\text{dis}} d_t}{n^{\text{dis}}}, \qquad t \in T \tag{72}$$

$$0 \le S_{t,s} \le \overline{S}, \qquad t \in T^{+} \tag{73}$$

$$0 \le P_{t,s}^{\text{ch}} \le \overline{P}^{\text{ch}} z_{t,s}^{\text{ch}}, \qquad 0 \le P_{t,s}^{\text{dis}} \le \overline{P}^{\text{dis}} z_{t,s}^{\text{dis}} \tag{74}$$

$$z_{t,s}^{\text{ch}} + z_{t,s}^{\text{dis}} \le 1, \qquad z_{t,s}^{\text{ch}}, z_{t,s}^{\text{dis}} \in \{0,1\} \tag{75}$$

To prevent cross-period arbitrage, total stored energy satisfies initial and terminal state constraints. The physical SOC adopts a hard terminal constraint because the storage energy state can be directly controlled by the PCS and BMS and serves as the handover condition for rolling dispatch. The low-carbon attribute inventory is affected by uncertainty in renewable energy and green-attribute arrival; therefore, Section 4.3 uses a soft terminal constraint.

$$S_{1,s} = S^{\text{ini}}, \qquad S_{N_T+1,s} = S^{\text{ter}}, \qquad s \in S \tag{76}$$

Cooling-system power is jointly determined by heat load and cooling

Table 1
Objective-function weights and penalty coefficients.

Parameter	Value	Unit	Description
Grid electricity price	0.33 / 0.62 / 0.92	CNY/kWh	Valley/flat/peak electricity price
Low-carbon attribute premium	60	CNY/MWh	Premium for externally purchased low-carbon electricity
Storage cycling cost	50	CNY/MWh throughput	Charging and discharging degradation cost
Renewable-curtailment penalty	20	CNY/MWh	Penalty for curtailed renewable electricity
Park-emission coefficient	100	CNY/tCO_2	Monetary-equivalent cost of park-level market-based emissions
Expected label-violation coefficient	1000	CNY/(kgCO_2 per workload unit)	Penalty for expected task-label violation
SLA-shortfall coefficient	1000	CNY/workload unit	Penalty for uncompleted workload
Label-CVaR coefficient	2000	CNY/(kgCO_2 per workload unit)	Penalty for tail label-violation risk
Checkpoint-action cost	200	CNY/action	Cost of a restart or migration-related checkpoint action
Cooling-capacity slack penalty	100,000	CNY/(MW·period)	Penalty for exceeding zonal cooling capacity
Terminal low-carbon-inventory penalty	10,000	CNY/MWh	Penalty for terminal low-carbon-inventory shortfall
Task SLA weight	1	Dimensionless	Relative weight of task SLA shortfall
Task label weight	1	Dimensionless	Relative weight of task label

efficiency. To ensure physical closure of cooling, the background heat load that is not allocated to specific tasks is first defined as

$$Q^{\mathrm{heat}}_{0,z,t} = Q^{\mathrm{env}}_{z,t} + Q^{\mathrm{aux,heat}}_{z,t} + \sum_{g:z(g)=z} k^{\mathrm{heat}}_{g} P^{\mathrm{idle}}_{g} y_{g,t} - \sum_{j\in J}\sum_{g:z(g)=z} k^{\mathrm{heat}}_{g} f^{\mathrm{idle}}_{j,g,t} P^{\mathrm{idle}}_{g} x_{j,g,t} \tag{77}$$

and $Q^{\mathrm{heat}}_{0,z,t} \geq 0$ is required. Here, $Q^{\mathrm{aux,heat}}_{z,t}$ explicitly receives the heat load generated by network, storage, power-distribution, and other auxiliary equipment and entering the computer-room or equipment-room cooling boundary. If these devices have independent heat-dissipation systems, this term can be set to zero according to actual conditions or another cooling zone can be modeled. The historical thermal response of cooling zone z consists of the task-resolved carry-over terms and the background carry-over term.

$$Q^{\mathrm{hist},k}_{z,t} = \sum_{j\in \mathscr{J}^{\mathrm{led},k}} Q^{\mathrm{hist},k}_{j,z,t} + Q^{\mathrm{hist},k}_{0,z,t} \tag{78}$$

where $\mathscr{J}^{\mathrm{led},k}$ includes both executable tasks and completed tasks with unsettled thermal tails. The total response heat load of cooling zone z is

$$Q^{\mathrm{heat},k}_{z,t} = Q^{\mathrm{hist},k}_{z,t} + \sum_{l=0}^{\min\{H_z,t-1\}} m^{\mathrm{th}}_{z,l}\left(\sum_{j\in \mathscr{J}^{\mathrm{led},k}} Q^{\mathrm{heat},k}_{j,z,t-l} + Q^{\mathrm{heat},k}_{0,z,t-l}\right) \tag{79}$$

The historical term is fixed at the beginning of each rolling window using realized PDU and cooling-meter records. It is updated after the implemented period and transferred to the next rolling window. The zonal cooling power satisfies

$$P^{\mathrm{cool}}_{z,t,s} = \frac{\widetilde{Q}^{\mathrm{heat}}_{z,t}}{\mathrm{COP}_{z,t,s}} \tag{80}$$

$$0 \leq P^{\mathrm{cool}}_{z,t,s} \leq \overline{P}^{\mathrm{cool}}_{z} + v^{\mathrm{cool}}_{z,t,s}, \qquad v^{\mathrm{cool}}_{z,t,s} \geq 0 \tag{81}$$

$$P^{\mathrm{cool}}_{t,s} = \sum_{z\in Z} P^{\mathrm{cool}}_{z,t,s} \tag{82}$$

4.3. Unique allocation of low-carbon attributes, task labels, and tail-risk constraints

The traceable energy of task j in period t under scenario s is

$$E^{\mathrm{trace}}_{j,t,s} = E^{\mathrm{IT}}_{j,t} + E^{\mathrm{cool}}_{j,t,s} + E^{\mathrm{sto}}_{j,t,s} + E^{\mathrm{net}}_{j,t} \tag{83}$$

where the IT energy is given by (28); the scenario-dependent cooling energy is specifically

$$E^{\mathrm{cool},k}_{j,t,s} = \Delta t \sum_{z\in Z} \frac{Q^{\mathrm{hist},k}_{j,z,t} + \sum_{l=0}^{\min\{H_z,t-1\}} m^{\mathrm{th}}_{z,l} Q^{\mathrm{heat},k}_{j,z,t-l}}{COP_{z,t,s}} \tag{84}$$

Storage-loss allocation follows the fixed traceability rule, and $\overline{b}^{\mathrm{sto}}_{j,t}$ is required to satisfy (45):

$$0 \leq E^{\mathrm{sto}}_{j,t,s} \leq \overline{b}^{\mathrm{sto}}_{j,t} E^{\mathrm{sto,loss}}_{t,s}, \qquad E^{\mathrm{sto}}_{j,t,s} \leq M^{\mathrm{sto}}_{t} b_{j,t} \tag{85}$$

$$E^{\mathrm{sto}}_{j,t,s} \geq \overline{b}^{\mathrm{sto}}_{j,t} E^{\mathrm{sto,loss}}_{t,s} - M^{\mathrm{sto}}_{t}\left(1 - b_{j,t}\right), \qquad \sum_{j\in J} E^{\mathrm{sto}}_{j,t,s} \leq E^{\mathrm{sto,loss}}_{t,s} \tag{86}$$

The park-metered energy is

$$E^{\mathrm{meter}}_{t,s} = \left(P^{\mathrm{IT}}_{t} + P^{\mathrm{cool}}_{t,s} + P^{\mathrm{aux}}_{t}\right) d_t + E^{\mathrm{sto,loss}}_{t,s} \tag{87}$$

and the energy-ledger consistency constraint is satisfied:

$$\sum_{j\in J} E^{\mathrm{trace}}_{j,t,s} + E^{\mathrm{trace}}_{0,t,s} = E^{\mathrm{meter}}_{t,s}, \qquad E^{\mathrm{trace}}_{0,t,s} \geq 0 \tag{88}$$

The low-carbon electricity entering the low-carbon ledger under scenario s is

$$P^{\mathrm{cool},k}_{t,s}\Delta t = \sum_{j\in \mathscr{J}^{\mathrm{led},k}} E^{\mathrm{cool},k}_{j,t,s} + E^{\mathrm{cool},k}_{0,t,s} \tag{89}$$

The above equality closes the cooling-energy ledger in each period. Task-level cooling electricity and background cooling electricity jointly recover the metered cooling electricity, including the thermal response inherited from preceding rolling windows. Consequently, no cooling electricity is omitted or transferred between tasks because of the rolling-window boundary.

$$E^{\mathrm{low,in}}_{t,s} = P^{\mathrm{ren,use}}_{t,s} d_t + E^{\mathrm{low,grid}}_{t,s} \tag{90}$$

Externally purchased low-carbon electricity satisfies

$$0 \leq E^{\mathrm{low,grid}}_{t,s} \leq \overline{E}^{\mathrm{low,grid}}_{t,s}, \qquad E^{\mathrm{low,grid}}_{t,s} \leq P^{\mathrm{grid}}_{t,s} d_t \tag{91}$$

The low-carbon electricity available for task claims is

$$E^{\mathrm{low,avl}}_{t,s} = E^{\mathrm{low,in}}_{t,s} - E^{\mathrm{low,ch}}_{t,s} + E^{\mathrm{low,dis}}_{t,s} \tag{92}$$

The low-carbon energy inventory in storage satisfies

$$S^{\mathrm{low}}_{t+1,s} = S^{\mathrm{low}}_{t,s} + n^{\mathrm{ch}} E^{\mathrm{low,ch}}_{t,s} - \frac{E^{\mathrm{low,dis}}_{t,s}}{n^{\mathrm{dis}}}, \qquad t \in T \tag{93}$$

$$0 \leq E^{\mathrm{low,ch}}_{t,s} \leq E^{\mathrm{low,in}}_{t,s}, \qquad 0 \leq E^{\mathrm{low,ch}}_{t,s} \leq P^{\mathrm{ch}}_{t,s} d_t \tag{94}$$

$$0 \leq E^{\mathrm{low,dis}}_{t,s} \leq P^{\mathrm{dis}}_{t,s} d_t, \qquad 0 \leq E^{\mathrm{low,dis}}_{t,s} \leq n^{\mathrm{dis}} S^{\mathrm{low}}_{t,s} \tag{95}$$

$$0 \leq S^{\mathrm{low}}_{t,s} \leq S_{t,s}, \qquad t \in T^{+} \tag{96}$$

The initial and terminal states of the low-carbon inventory adopt a

soft terminal constraint:

$$S^{\mathrm{low}}_{1,s} = S^{\mathrm{low,ini}}, \qquad S^{\mathrm{low}}_{N_T+1,s} + v^{\mathrm{low,ter}}_{s} \geq S^{\mathrm{low,ter}}, \qquad v^{\mathrm{low,ter}}_{s} \geq 0 \tag{97}$$

and the parameters themselves are required to satisfy

$$0 \leq S^{\mathrm{low,ini}} \leq S^{\mathrm{ini}}, \qquad 0 \leq S^{\mathrm{low,ter}} \leq S^{\mathrm{ter}} \tag{98}$$

Eq. (97) avoids infeasibility caused by forcing the low-carbon inventory to return exactly to a fixed value under scenarios with low renewable output or insufficient arrival of low-carbon attributes. The terminal shortfall is recorded by $v^{\mathrm{low,ter}}_{s}$ and penalized in the objective function. The physical SOC does not use a soft terminal constraint because it reflects the true battery energy state and must provide an executable initial value for the next rolling window. The low-carbon inventory reflects an attribute ledger, allowing deviations to be recorded and settled through penalty terms.

In practice, the energy-management system can track the attribute inventory in a logical ledger alongside the battery SOC. Each eligible charging event creates a timestamped batch that records the charging interval, PCS meter identifier, low-carbon electricity source or transaction identifier, emission factor, efficiency-adjusted eligible quantity, and unclaimed balance. When a storage-backed low-carbon attribute is released, the system deducts the corresponding quantity from the relevant batch, links it to a unique task-claim identifier, and retires it to prevent reuse. The PCS measurements, transaction or certificate records, inventory balances, and retirement records provide the evidence required for ex post verification.

The allocation of claimable low-carbon electricity satisfies

$$0 \leq A^{\mathrm{low}}_{j,t,s} \leq E^{\mathrm{trace}}_{j,t,s}, \qquad 0 \leq A^{\mathrm{low}}_{0,t,s} \leq E^{\mathrm{trace}}_{0,t,s} \tag{99}$$

$$\sum_{j\in J} A^{\mathrm{low}}_{j,t,s} + A^{\mathrm{low}}_{0,t,s} \leq E^{\mathrm{low,avl}}_{t,s}, \qquad t \in T,\ s \in S \tag{100}$$

Eq. (99) means that a task can obtain a low-carbon claim only when it is actually executed and forms traceable energy. Eq. (100) means that the low-carbon electricity in the same period can be allocated only once.

The task-level carbon labels and carbon-reduction results reported in this paper follow a market-based Scope 2 accounting perspective. Under this perspective, the carbon emissions assigned to a task depend on its traceable electricity consumption and the associated electricity attributes. Electricity covered by a unique low-carbon claim is evaluated using the low-carbon claim emission factor, whereas the remaining electricity is evaluated using the residual-mix emission factor. These attributes provide accounting evidence for electricity consumption rather than describing the physical routing of electricity, and their allocation does not alter the park power-balance constraints. As expressed in (109), task-level emissions and background emissions are accounted for within the same boundary and together form the park-level market-based emissions.

The location-based perspective provides a separate measure of emissions associated with the electricity physically purchased from the grid. It applies the grid-average emission factor to grid purchases and does not recognize contractual low-carbon electricity attributes or their allocation among individual tasks. Unless otherwise stated, the task carbon intensities, label-achievement rates, and park-level carbon-reduction percentages reported in this paper are all market-based results. The location-based metric is used only as a separate physical-emission indicator and does not determine task-level low-carbon labels.

Under the above constraints, the traceable carbon emissions of task j under scenario s are

$$C^{\mathrm{trace}}_{j,s} = \sum_{t\in T}\left[EF^{\mathrm{low}}_{t,s} A^{\mathrm{low}}_{j,t,s} + EF^{\mathrm{res}}_{t,s}\left(E^{\mathrm{trace}}_{j,t,s} - A^{\mathrm{low}}_{j,t,s}\right)\right] \tag{101}$$

The background residual carbon emissions are

$$C^{\mathrm{trace}}_{0,s} = \sum_{t\in T}\left[EF^{\mathrm{low}}_{t,s} A^{\mathrm{low}}_{0,t,s} + EF^{\mathrm{res}}_{t,s}\left(E^{\mathrm{trace}}_{0,t,s} - A^{\mathrm{low}}_{0,t,s}\right)\right] \tag{102}$$

The task-level carbon intensity is

$$CI_{j,s} = \frac{C^{\mathrm{trace}}_{j,s}}{W_j}, \qquad j \in J^{\mathrm{low}} \tag{103}$$

Because (25) already requires $s^{\mathrm{SLA}}_{j} = 0$ for low-carbon label tasks, the problem of a task not being completed but obtaining a falsely favorable carbon intensity due to a fixed denominator will not occur. For non-label tasks, only service shortfall and actual completed amount may be counted, and no formal low-carbon label is issued.

Under rolling implementation, task-level traceable energy, low-carbon electricity claims, and carbon emissions are accumulated only for implemented periods. Completion of the computational workload does not immediately close the task ledger if a nonzero thermal tail remains. The final carbon intensity of task j is calculated, and its low-carbon label is issued, only after $s^{\mathrm{SLA}}_{j} = 0$ and all task-resolved historical thermal-response terms have been settled. This rule ensures that cooling electricity consumed after computational completion remains included in the carbon responsibility of the task that generated the corresponding heat load.

For tasks applying for low-carbon computing services, the task-level carbon-label constraint should be satisfied:

$$CI_{j,s} \leq \overline{CI}_{j} + v^{\mathrm{tag}}_{j,s}, \qquad v^{\mathrm{tag}}_{j,s} \geq 0, \qquad j \in J^{\mathrm{low}},\ s \in S \tag{104}$$

If $v^{\mathrm{tag}}_{j,s} > 0$, task j has not fully satisfied its low-carbon label commitment under scenario s.

To further control the tail violation risk of low-carbon labels under adverse scenarios, the label-violation loss under scenario s is defined as

$$V^{\mathrm{tag}}_{s} = \sum_{j\in J^{\mathrm{low}}} w^{\mathrm{tag}}_{j} v^{\mathrm{tag}}_{j,s} \tag{105}$$

Given the confidence level $\alpha^{\mathrm{cv}} \in (0,1)$, the conditional value-at-risk of task-level low-carbon label-violation loss is defined as

$$R^{\mathrm{tag}} = \xi^{\mathrm{tag}} + \frac{1}{1-\alpha^{\mathrm{cv}}}\sum_{s\in S} p_s v^{\mathrm{tail}}_{s}, \qquad \xi^{\mathrm{tag}} \in \mathbb{R} \tag{106}$$

$$v^{\mathrm{tail}}_{s} \geq V^{\mathrm{tag}}_{s} - \xi^{\mathrm{tag}}, \qquad v^{\mathrm{tail}}_{s} \geq 0, \qquad s \in S \tag{107}$$

Further impose

$$R^{\mathrm{tag}} \leq \overline{R}^{\mathrm{tag}} \tag{108}$$

Eqs. (106)–(108) play a role different from that of the expected violation penalty in the objective function: the expectation term characterizes the average service level, whereas the CVaR hard constraint limits tail risk. The two respectively constrain average performance and label violation under low-probability adverse scenarios.

Finally, park carbon-emission consistency is expressed as follows:

$$\sum_{j\in J} C^{\mathrm{trace}}_{j,s} + C^{\mathrm{trace}}_{0,s} = C^{\mathrm{park}}_{s}, \qquad s \in S \tag{109}$$

This equation is a carbon-accounting definition rather than an independent physical constraint. The optimization feasible region is mainly defined by power balance, low-carbon inventory in storage, uniqueness of low-carbon claims, and task-label constraints. Thus, task-level carbon traceability has been embedded into the decarbonization dispatch model. In addition to satisfying task completion, computing-resource capacity, park power balance, cooling-capacity boundary, and storage-operation constraints, a feasible solution of the model must also satisfy task traceable-energy consistency, unique low-carbon electricity claims, task-level carbon labels, and label tail-risk constraints.

From the model structure, when $\mathrm{COP}_{z,t,s}$, thermal response kernel coefficients, traceability allocation coefficients, and scenario parameters

are given, the above model can be transformed into a mixed-integer linear programming model. If temperature-dependent $COP_{z,t,s}$ or endogenous identification of proportional allocation coefficients is further considered, the model becomes a mixed-integer nonlinear programming model. In engineering implementation, rolling optimization can be used: first, a task plan is formed according to short-term forecasts of task arrivals, carbon factors, renewable output, and trace cooling states; then $E^{\text{trace}}_{j,t,s}$, $A^{\text{low}}_{j,t,s}$, and $CI_{j,s}$ are updated according to actual task logs, energy-consumption metering, and low-carbon electricity records, and the next dispatch cycle begins. The detailed record-matching and ex-post verification procedure is described in Section 5.1.3.

5. Case studies

5.1. Case setup

5.1.1. Test systems and data settings

Two test systems are used to verify the proposed task-level carbon traceability and decarbonization dispatch method. The first system is the modified Yunnan Chenggong park distribution network, and the second system is the modified IEEE 118-bus system. The topology of the Yunnan Chenggong park system is shown in Fig. 3, and the main parameters of the park energy system are listed in Table 2.

Representative two-week photovoltaic and wind generation profiles for the Chenggong site were derived from the NASA POWER hourly solar radiation and wind speed datasets [39]. The baseline capacity factors were set to 0.155 for solar and 0.290 for wind based on regional historical benchmarks. Additionally, the time-of-use electricity tariff structure aligns with Yunnan's local pricing policy [40], with rounded peak, flat, and valley rates of 0.92, 0.62, and 0.33 CNY/kWh, respectively.

According to the renewable capacities and capacity factors in Table 2, the two-week available renewable energy of the park under the baseline scenario is 3251 MWh. After further combining the scenario probabilities in Table 7, the scenario-weighted expected available renewable energy is 2910 MWh.

The modified IEEE 118-bus system retains the original system network topology, conventional loads, conventional generators, and line parameters, and connects three AIDC parks in different load areas. The access settings of each park are shown in Table 3. According to the installed capacities in the table and the same capacity factors as the Chenggong park, the baseline two-week available renewable energy of the three parks is 9777 MWh in total, and the scenario-weighted expectation is 8750 MWh. Dispatch is performed in a rolling manner, with a rolling window of 24 h, a time interval of 1 h, and continuous rolling for 14 days, totaling 336 periods.

Table 2
Energy-system parameter settings of the Chenggong park.

Parameter	Value
Peak IT power	30 MW, including a 12 MW training pool, 10.5 MW general pool, and 7.5 MW inference pool
Storage configuration	20 MW / 80 MWh
Local renewable capacity	PV: 25 MW, WT: 20 MW
Two-week average capacity factor of PV/WT	0.155 / 0.290
Time-of-use electricity price, peak/flat/valley	0.92 / 0.62 / 0.33 CNY/kWh
Low-carbon attribute premium $c^{\text{low,grid}}$	60 CNY/MWh
Upper limit of externally purchased low-carbon electricity	25% of peak facility load, about 10 MWh/h

Table 3
AIDC park access settings in the modified IEEE 118-bus system.

Park	Access bus	Peak IT power/ MW	Typical PUE	Storage configuration/ MW-MWh	Local renewable configuration/ MW
AIDC-1	Bus 33	28	1.33	18–72	PV: 22, WT: 18
AIDC-2	Bus 52	36	1.33	24–96	PV: 26, WT: 24
AIDC-3	Bus 83	30	1.33	20–80	PV: 20, WT: 22

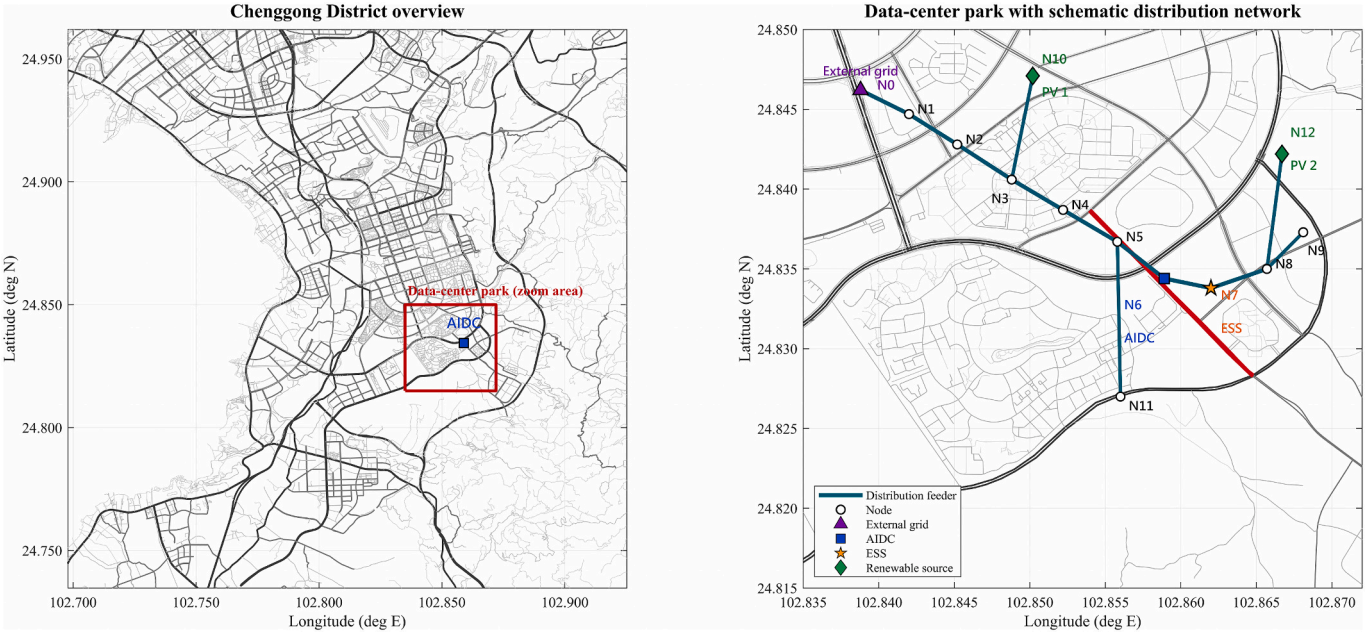


Fig. 3. Location and equivalent distribution-network topology of the AI computing center park in Chenggong District.

For AIDC park i connected to bus $b(i)$, its equivalent active load is modified as

$$P^{\text{load,new}}_{b(i),t,s} = P^{\text{load,base}}_{b(i),t} + P^{\text{IT}}_{i,t} + P^{\text{cool}}_{i,t,s} + P^{\text{aux}}_{i,t} + P^{\text{ch}}_{i,t,s} - P^{\text{dis}}_{i,t,s} - P^{\text{ren,use}}_{i,t,s} \quad (110)$$

Eq. (110) provides the hourly net active-power demand of each AIDC park at its point of connection. The park-side scheduling model and the external network calculation are performed sequentially. After each rolling-window solution, the resulting net demands of AIDC-1, AIDC-2, and AIDC-3 are added to buses 33, 52, and 83, respectively, while the remaining loads and generator data follow the modified IEEE 118-bus system. The reactive-power demand of each AIDC park is represented using a constant lagging power factor of 0.95.

An AC optimal power flow is then performed for each hour and operating scenario to verify network feasibility. The calculation considers the active- and reactive-power limits of conventional generators, bus-voltage limits of 0.95–1.05 p.u., the reference-bus condition, and branch apparent-power limits. The external network calculation serves as an ex-post feasibility check and is not embedded in the task–energy scheduling MILP.

5.1.2. Task, resource-pool, and low-carbon traceability parameters

Each AIDC park contains three types of computing resource pools, corresponding to the high-performance training resource pool, the general training and fine-tuning resource pool, and the batch inference resource pool. The IT power shares of the three resource pools are 40%, 35%, and 25%, respectively (See Table 4). The base power is set to 15% of the full-load IT power, and the remaining power varies with the task computing-service rate. Let the full-load IT power of resource pool g be $\overline{P}^{\text{IT}}_g$ and the conversion coefficient be ρ_g. Then

$$R_g = \rho_g \overline{P}^{\text{IT}}_g, \qquad P^{\text{idle}}_g = 0.15\overline{P}^{\text{IT}}_g, \qquad k^{\text{it}}_{j,g} = \frac{0.85\overline{P}^{\text{IT}}_g}{R_g} \quad (111)$$

From (111), the dynamic IT energy per unit workload of the three resource pools is 53.1, 47.2, and 35.4 kWh/unit, respectively. After considering the allocation of base power consumption, cooling electricity consumption, and network-storage energy consumption, the facility-scope unit energy consumption of the three types of tasks is approximately 78.5, 70.5, and 52.8 kWh/unit, respectively, with the cooling-system COP ranging from 3.8 to 4.5.

The dispatchable task set of each AIDC park over the 14-day rolling cycle is shown in Table 5, and the Chenggong park adopts the same single-park task set. Thus, the Chenggong park contains 39 dispatchable tasks, among which 20 apply for low-carbon labels. The modified IEEE 118-bus system contains 117 dispatchable tasks across three parks, among which 60 apply for low-carbon labels. Online inference requests have strong real-time requirements and are aggregated as non-transferable baseline loads in this paper, rather than participating in cross-period task scheduling. Baseline loads and background energy consumption account for approximately 81%–82% of total park energy use, whereas dispatchable task energy consumption accounts for approximately 18%–19%.

Table 4
Equivalent parameters of AIDC resource pools.

Resource-pool type	Applicable tasks	IT power share	Conversion coefficient ρ_g/unit MW^{-1} h^{-1}
High-performance training pool	Large-model training and long-duration fine-tuning	40%	16.0
General training pool	Small- and medium-scale model training and parameter fine-tuning	35%	18.0
Batch inference pool	Offline inference and batch-processing tasks	25%	24.0

Table 5
AI task parameter settings of each AIDC park.

Task type	Quantity	Workload range/unit	Executable window/h	Number of label-applying tasks
Large-model training tasks	9	1000–1800	12–18	6
Model fine-tuning tasks	12	350–800	6–12	7
Batch inference tasks	18	120–350	3–8	7

The range of the grid carbon emission factor was calibrated against China's official 2022 electricity CO_2 emission factors, which specify 0.5366 kgCO_2/kWh for the national average, 0.5856 kgCO_2/kWh for the national average excluding market-traded non-fossil electricity, and 0.8325 kgCO_2/kWh for fossil-based generation [41]. These benchmark values establish the reference bounds for the adopted range of EFgrid t, $s = 0.45$–0.82 kgCO_2/kWh. The treatment of the residual-mix factor follows the GHG Protocol Scope 2 Guidance [42]. Due to the unavailability of public time-resolved residual-mix datasets for the study area, a representative setting of $EF^{res}_{t,s} = 0.42$–0.78 kgCO_2/kWh (with an expected value of 0.62 kgCO_2/kWh) was adopted for the case study.

The decarbonization dispatch and carbon traceability parameters are shown in Table 6. The task-level labels directly adopt the four levels A/B/C/D defined in (60), and the thresholds B_1, B_2, B_3 are 5, 15, and 30 kgCO_2 /unit workload, respectively. Label achievement is defined as $CI_j \le \overline{CI}_j = 15$, namely reaching level A or B. The task carbon intensity and label grade in the main result tables both use the expected-carbon-intensity perspective, i.e., $CI^{\text{exp}}_j = \sum_s p_s CI_{j,s}$. The CVaR confidence level is set to $\alpha^{\text{cv}} = 0.90$. Because the scenario set contains five discrete scenarios and the minimum scenario probability is 0.10, $1 - \alpha^{\text{cv}} = 0.10$ matches the tail-risk measure with the resolution of scenario probabilities.

The cooling system is modeled by park zones. Each AIDC park has two cooling zones. The training resource pool and the general resource pool are connected to the high-density cooling zone, whereas the batch inference resource pool is connected to the ordinary cooling zone.

Under normal scenarios, the COP is set to 3.8–4.5. This range was selected with reference to U.S. Department of Energy and Oak Ridge National Laboratory data-center cooling benchmarks, which report a COP of 3.8 for high-efficiency rooftop cooling and 4.6 for a typical installed centrifugal chiller [43]. Under high-temperature or cooling-efficiency degradation scenarios, the COP is reduced by 8%–12% as a stress-test setting.

The thermal response kernel is set as

Table 6
Decarbonization dispatch and carbon traceability parameters.

Parameter	Value or setting method
$\eta^{\text{ch}}, \eta^{\text{dis}}$	0.95, 0.95
Initial and terminal SOC of storage	50%, restored to 50% at the terminal
$S^{\text{low,ini}}$	30% of initial stored energy
$EF^{\text{low}}_{t,s}$	0.03–0.08 kgCO_2/kWh, expected value 0.05
$EF^{\text{res}}_{t,s}$	0.42–0.78 kgCO_2/kWh, expected value 0.62
$EF^{\text{grid}}_{t,s}$	0.45–0.82 kgCO_2/kWh
B_1, B_2, B_3	5, 15, 30 kgCO_2/unit workload
$\overline{CI}_j$	15 kgCO_2/unit workload
w^{tag}_j	1
α^{cv} $\overline{R}^{\text{tag}}$	0.90, 6.0 kgCO_2/unit workload

$$m^{\mathrm{th}}_{z,0} = 0.70, \quad m^{\mathrm{th}}_{z,1} = 0.20, \quad m^{\mathrm{th}}_{z,2} = 0.08, \quad m^{\mathrm{th}}_{z,3} = 0.02 \tag{112}$$

with $H_z = 3$ and

$$\sum_{l=0}^{H_z} m^{\mathrm{th}}_{z,l} = 1 \tag{113}$$

The final coefficient represents the lumped residual response beyond the first three modeled components. This residual response is carried across rolling windows with the identity of the task that generated it. After the final operating period of the 14-day simulation, three settlement-only periods are appended to clear the remaining thermal-response states; no new task execution is permitted during these periods.

5.1.3. Operating scenarios, comparison methods, and ex-post audit protocol

To characterize uncertainty in renewable generation, residual carbon factors, and cooling efficiency, this paper constructs five operating scenarios, as shown in Table 7. The renewable output coefficient acts simultaneously on the typical local photovoltaic P and wind output curves. The scenario probabilities satisfy $\sum_{s\in S} p_s = 1$, and the expected output coefficient is 0.895.

Four comparison methods are set, as shown in Table 8. M1 is the baseline operating mode without low-carbon claim optimization, M2 is park-level aggregate decarbonization dispatch, M3 is a task-level allocation method without low-carbon inventory in storage, and M4 is the complete method proposed in this paper.

For an ex-post audit, records are aligned with the 1 h dispatch intervals using task identifiers, resource-pool identifiers, meter mappings, and timestamps. Platform logs reconstruct each task's execution history, including its resource-pool assignment, interruptions, migrations, and completion status. Server telemetry and rack-PDU records provide the task-level IT energy data. Measurements from the cooling branches, network and storage equipment, and storage PCS are allocated under the fixed traceability rules and reconciled with the park-boundary meter. Any energy that cannot be reliably attributed to a task remains in the background ledger.

Local renewable measurements, hourly purchased-low-carbon records, and eligible storage-attribute batches are matched to $A^{\mathrm{low}}_{j,t}$. A quantity is accepted only when it has a supporting source record and a unique claim identifier that has not been used previously. The audited records are used to recalculate $E^{\mathrm{trace}}_{j,t}$, C^{trace}_{j}, and CI_j.

Verification requires the task to be complete and its thermal tail to be settled. The task and park energy ledgers must close within the adopted metering tolerance. In addition, the low-carbon allocation must be unique and supported by the inventory, and the resulting carbon intensity must meet the committed threshold. If any of these conditions is not met, the claim is rejected or downgraded, and the discrepancy remains in the audit record.

For empirical validation, the three ledgers would be reconstructed from independently recorded operational data. Platform task logs would provide task identifiers, arrival and completion times, executed workloads, resource-pool assignments, interruptions, and migrations. Server telemetry and rack-PDU measurements would be used to compare the modeled task-level IT energy with measured server energy, while cooling-branch, network-storage, park-boundary, and storage-PCS meters would be used to test subsystem and park-level energy-ledger closure. Hourly low-carbon electricity transaction records, renewable-meter data, and certificate-retirement records would then be used to verify whether each modeled low-carbon allocation was available in the corresponding period and had not been claimed previously. Validation metrics would include the task-energy estimation error, subsystem and park energy-balance residuals, the amount of unmatched or duplicate low-carbon claims, and agreement between model-derived and audited task labels.

Table 7
Operating scenario settings.

Scenario	Probability	Renewable output coefficient	Residual carbon factor variation	Cooling COP variation
S1: Normal operation	0.45	1.00	Baseline value	Baseline value
S2: Low renewable output	0.20	0.75	Baseline value	Baseline value
S3: Increased residual carbon factor	0.15	0.90	+20%	Baseline value
S4: Reduced cooling efficiency	0.10	0.95	Baseline value	−10%
S5: Adverse multi-factor superposition	0.10	0.65	+25%	−12%

Table 8
Comparison method settings.

Method	Description
M1	Tasks are fixedly scheduled at the earliest feasible time. Storage, renewable energy, and grid purchases are dispatched to minimize operating cost. Low-carbon electricity claim optimization is not performed.
M2	Park-level decarbonization dispatch with task time shifting, where total market-based emissions are included as one optimization objective. Low-carbon electricity is not uniquely allocated at the task level.
M3	Task-level carbon label constraints and task-level allocation of low-carbon electricity are considered, but no low-carbon attribute inventory for storage is established; green-electricity attributes cannot be carried across periods through storage.
M4	Proposed method: task-level carbon traceability, unique low-carbon electricity claims, low-carbon inventory in storage, and label CVaR risk constraints are enabled simultaneously.

5.2. Task-energy coordinated dispatch result analysis

Fig. 4 and Fig. 5 show the typical two-week dispatch results of the Chenggong park under the proposed method. The figure shows that the IT load of the AIDC rises during the midday periods with high renewable output on most days and decreases during evening peak periods. Cooling power varies jointly with IT load and period-specific COP. Because some tasks are shifted to high-temperature midday periods, the average PUE of M2–M4 increases from 1.332 in M1 to 1.334–1.336. This change indicates that although task time shifting improves low-carbon electricity matching, it also brings a slight cooling cost; decarbonization dispatch is therefore not cost-free for all operating metrics.

Table 9 gives the scenario-expected dispatch metrics of the Chenggong park under different methods. For each method, grid-purchased electricity, renewable utilized electricity, storage losses, and facility electricity consumption satisfy the energy-closure relationship. Taking M4 as an example, $6155.8 + 2883.8 = 8998.0 + 41.6$ MWh, indicating that the sum of purchased electricity and renewable utilization equals the sum of facility electricity consumption and storage losses. Compared with M1, the operating cost of M4 decreases from 4.065 million CNY to 3.797 million CNY, a reduction of 6.6%; market-based emissions decrease from 4009 t to 3556 t, a reduction of 11.3%; and peak grid purchase decreases from 31.52 MW to 26.08 MW. The results show that the proposed method improves operating cost, carbon emissions, and peak grid purchase while maintaining energy closure.

M4 is not superior to M3 in all economic operating metrics. First, the operating cost of M4 is 18 thousand CNY higher than that of M3, corresponding to an increase of 0.48%. This is because M4 purchases an additional 710 MWh of low-carbon attribute electricity and increases storage throughput to satisfy label and tail-risk constraints. This reflects

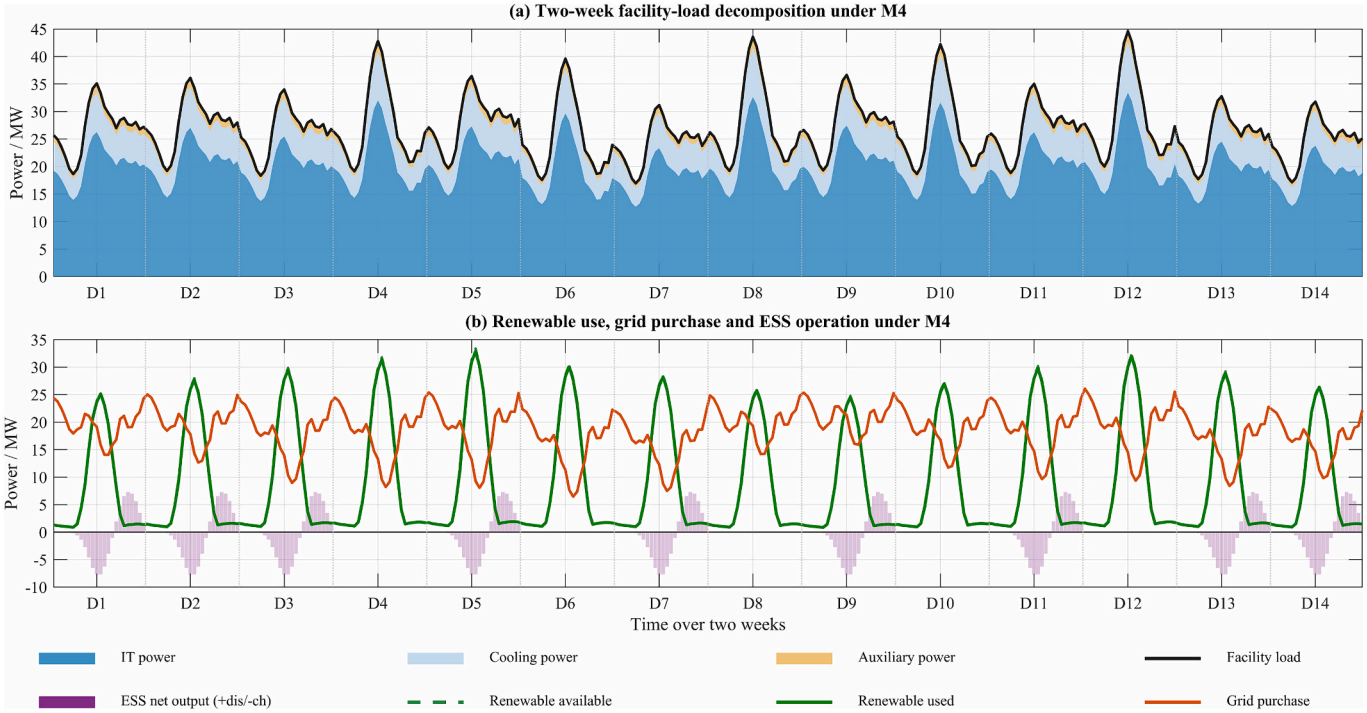


Fig. 4. Two-week task-energy coordinated dispatch results of the Chenggong park under the proposed method.

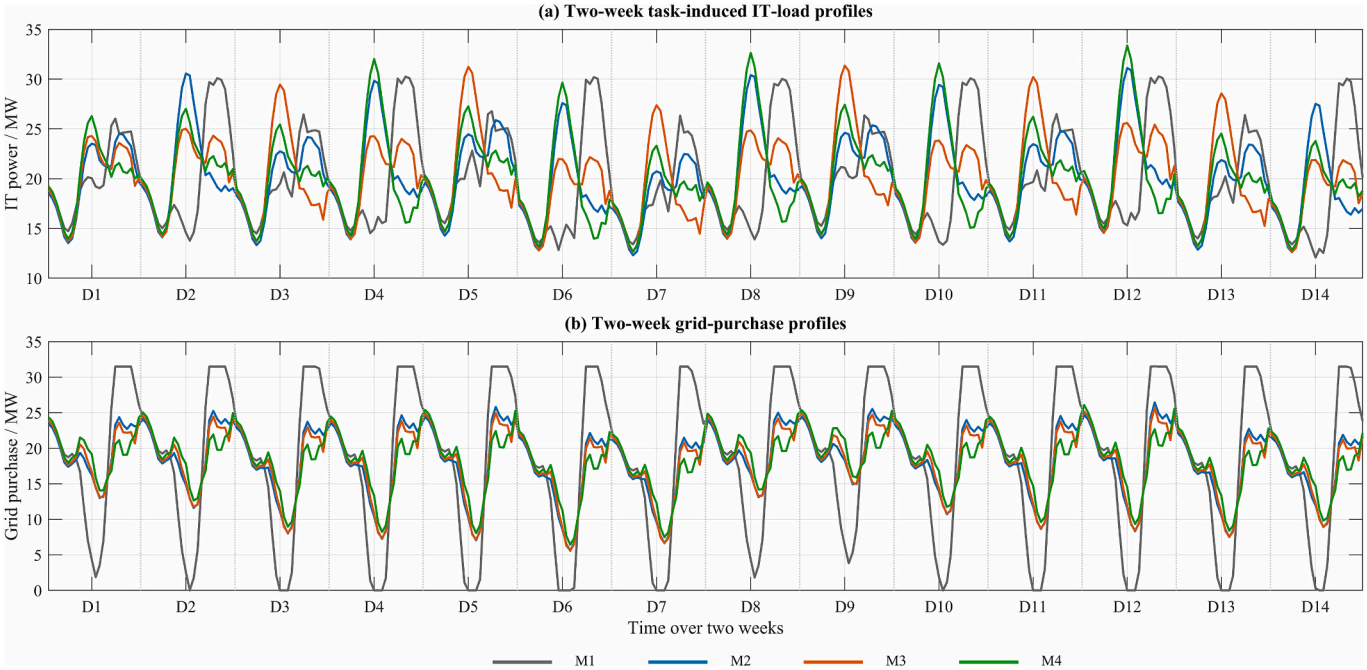


Fig. 5. Comparison of two-week IT load and grid electricity purchase curves of the Chenggong park under different methods.

Table 9
Comparison of two-week task-energy coordinated dispatch metrics of the Chenggong park.

Method	Operating cost/ 10^4 CNY	Grid purchase/ MWh	Peak purchase/MW	Renewable utilization rate/%	Renewable curtailment/MWh	Storage throughput/MWh	Average PUE	Market-based emissions/tCO_2
M1	406.5	6243.9	31.52	94.60	157.1	621.8	1.332	4009
M2	378.1	6132.4	26.44	98.90	32.0	693.5	1.334	3764
M3	377.9	6138.6	25.71	99.30	20.4	746.2	1.335	3654
M4	379.7	6155.8	26.08	99.10	26.2	812.7	1.336	3556

that carbon-performance improvement and risk control require a certain economic cost. Second, the renewable curtailment of M4 is 26.2 MWh, slightly higher than the 20.4 MWh of M3. This is because the storage-charging schedule in M4 serves not only electricity absorption but also low-carbon inventory construction, causing the pure electricity-optimal charging arrangement to be abandoned in a few periods. Finally, the peak grid purchase of M4 is 26.08 MW, also slightly higher than that of M3. The operating-cost differences between M3 and M4 are 0.48% for the Chenggong park and 0.46% for the modified IEEE 118-bus system, both exceeding the prescribed relative MIP gap of 0.1%. Their directions are also consistent with the additional low-carbon procurement, storage-throughput, and label-risk requirements of M4, indicating that the differences arise from the modeled cost–carbon–risk tradeoff rather than solver tolerance.

To further distinguish the sources of peak reduction, this paper conducts a control experiment in which the M4 task plan is fixed and storage is disabled. The peak grid purchase in this control scenario is 29.86 MW. Accordingly, M1 to M4 achieve a total peak reduction of 5.44 MW, of which task rescheduling contributes approximately 1.66 MW and storage discharge contributes approximately 3.78 MW. Thus, peak reduction is mainly undertaken by storage, while task time shifting plays an auxiliary role. This result is consistent with the system characteristic that the time-shifting margin of training tasks is limited. Because dispatchable task energy consumption accounts for only about 20% of total park energy use, its direct influence on the shape of the total load curve is limited; however, it has a significant effect on the time matching of low-carbon electricity and label achievement. The relevant results are further analyzed in Section 5.3.

The model was implemented in MATLAB R2023b and solved through the direct MATLAB interface of Gurobi 10.0.1 on an Intel Xeon workstation with 64 GB of memory. The relative MIP gap was set to 0.1% [44], the time limit was 300 s for each 24-h rolling window, and the number of solver threads was limited to 24. The default feasibility and integer-feasibility tolerances of 10^{-6} were retained. For the complete M4 formulation, the average solution time per rolling window was 9.2 s for the Chenggong park and 34.6 s for the modified IEEE 118-bus system, while the maximum solution times were 48.7 and 176.4 s, respectively. None of the reported rolling windows reached the time limit, and all solutions satisfied the prescribed MIP-gap criterion.

5.2.1. *Dispatch results of the modified IEEE 118-bus system*

Fig. 6 shows the aggregated dispatch results of the three AIDC parks in the modified IEEE 118-bus system, and Table 10 further provides a comparison of metrics under different methods.

The expected available renewable energy of the three parks over two weeks is 8750 MWh, corresponding to an integrated installed-capacity factor of approximately 0.21, which is consistent with typical photovoltaic and wind levels in Yunnan. Compared with M1, M4 reduces the operating cost by 6.4%, the aggregated market-based emissions of the three AIDC parks by 10.1%, and the peak grid purchase from 101.3 MW to 83.0 MW. The control experiment shows that task rescheduling contributes approximately 5.1 MW to the reduction in peak grid purchase, while storage contributes approximately 13.2 MW. This result is consistent with the Chenggong park case: storage mainly undertakes peak reduction, and task rescheduling mainly improves renewable-energy period matching and low-carbon attribute allocation.

Compared with M3, the operating cost of M4 is 56 thousand CNY higher, corresponding to an increase of 0.46%; renewable curtailment and peak grid purchase also increase slightly. However, the carbon emissions and subsequent task-level label metrics of M4 are significantly better than those of M3. Both systems exhibit a cost-carbon-risk tradeoff: the proposed method does not simply pursue the lowest operating cost, but improves the verifiability of task-level low-carbon claims and the robustness of labels under tail scenarios at a small economic cost. Across all reported hourly schedules, the bus-voltage magnitude ranged from 0.983 to 1.050 p.u., and the maximum branch loading was 88.1%. No

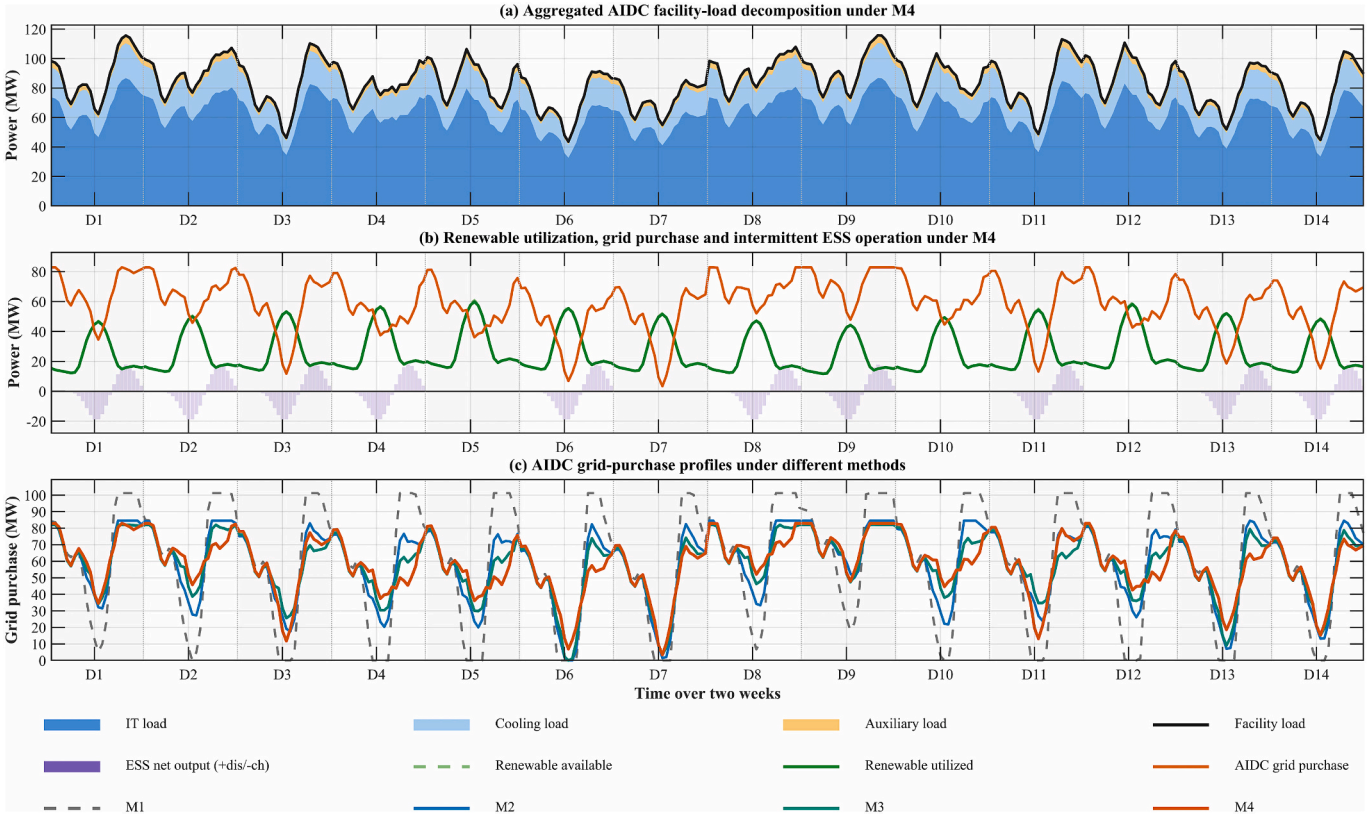


Fig. 6. Two-week task-energy coordinated dispatch results in the modified IEEE 118-bus system.

Table 10
Comparison of two-week dispatch metrics in the modified IEEE 118-bus system.

Method	Operating cost/ 10^4 CNY	AIDC grid purchase/MWh	Peak purchase/MW	Renewable utilization rate/%	Renewable curtailment/MWh	Storage throughput/MWh	Average PUE	Market-based emissions/tCO_2
M1	1298.8	20,027.1	101.3	93.80	542.5	1652	1.331	12,827
M2	1213.2	19,649.5	84.6	98.60	122.5	1894	1.333	12,141
M3	1209.9	19,629.9	82.1	99.20	70.0	2048	1.334	11,818
M4	1215.5	19,690.8	83.0	99.00	87.5	2213	1.334	11,530

voltage or branch-capacity violation was observed.

The park-emission coefficient was set to 100 CNY/tCO_2. At the reported M4 solutions, the corresponding carbon terms are approximately 0.356 million CNY for the Chenggong park and 1.153 million CNY for the modified IEEE 118-bus system, accounting for approximately 9.4%–9.5% of their physical operating costs. Carbon performance therefore affects the dispatch decisions without dominating the economic objective. The expected label-violation and label-CVaR coefficients were set to 1000 and 2000 CNY/($kgCO_2$ per workload unit), respectively, giving greater weight to low-probability adverse label outcomes than to average violations. All task weights were set to one because no additional priority differentiation was imposed among tasks of the same service class. The SLA-shortfall, cooling-capacity-slack, and terminal low-carbon-inventory penalties were selected to exceed the maximum economic benefit obtainable from relaxing the corresponding constraints. In particular, the maximum energy-cost saving from reducing one workload unit is below approximately 80 CNY under the adopted workload-energy coefficients and peak electricity price, which is substantially lower than the SLA-shortfall penalty of 1000 CNY per workload unit. Similarly, the cooling-capacity and terminal-inventory penalties are considerably higher than the marginal electricity and low-carbon-attribute costs. Consequently, cooling-capacity slack, SLA shortfall, and terminal low-carbon-inventory shortfall are zero in all reported main cases. Positive label-violation variables occur only for the one unachieved training task in the Chenggong park and the three unachieved training tasks in the modified IEEE 118-bus system.

5.3. Task-level carbon traceability and low-carbon label result analysis

This subsection turns to the task side and analyzes low-carbon electricity traceability and task-level label results. The Chenggong park contains 39 dispatchable tasks, of which 20 apply for labels. The modified IEEE 118-bus system contains 117 dispatchable tasks, of which 60 apply for labels. The above task scales are consistent with the settings in Table 5. In the tables, the low-carbon electricity claimed by tasks refers to the expected value of the low-carbon claimed electricity attributed to label-applying tasks after audit, i.e., $\sum_{j\in J^{\text{low}}}\sum_t A^{\text{low}}_{j,t}$. Label grades adopt the expected-carbon-intensity perspective.

5.3.1. Task-level carbon traceability and label results of the Chenggong Park

Table 11 and Fig. 7 give the carbon traceability and label results of the 20 label-applying tasks in the Chenggong park. The results are closed with the energy-accounting relationship. The two-week facility-scope energy consumption of the 20 label tasks is approximately 1030 MWh, and the claimed electricity of M1–M4 corresponds to low-carbon coverage rates of approximately 32%, 49%, 68%, and 84%, respectively. The equivalent emission factor calculated by coverage rate $c, EF^{\text{eff}} = 0.05c + 0.62\,(1-c)$, multiplied by the unit energy consumption of the three task types, 78.5, 70.5, and 52.8 kWh/unit, can reproduce the order of magnitude of the average carbon intensity in the table. Taking M4 as an example, the average coverage rates of training, fine-tuning, and inference tasks are approximately 83%, 82%, and 98%, respectively, corresponding to within-class average carbon intensities of approximately 11.6, 10.7, and 3.4 $kgCO_2$ /unit workload, and the task average is 8.4.

By method, M1 has the lowest low-carbon coverage because tasks are concentrated immediately after arrival and many tasks run at night and during evening peak periods. Under this method, all six label-applying training tasks fall into Grade D. M2 improves period matching through task time shifting, but park-average conversion produces 31.4 MWh of claim conflicts, accounting for about 6% of its claimed amount. Because M2 has no task-level directed allocation, the improvement in labels is limited. M3 significantly increases coverage after introducing task-level allocation, but because green-electricity attributes cannot be carried forward through storage, training task segments running at night cannot obtain low-carbon claims. Under this method, the coverage rate of training tasks is only about 62%, and the 41.2 MWh claim during storage-discharge periods is also removed by audit due to the lack of inventory evidence.

Through the low-carbon inventory of storage, M4 directionally releases the green-electricity attributes charged during the daytime to nighttime training periods, increasing the training coverage rate from about 62% to about 83%. This is the main source of the improvement in label achievement rate of M4 relative to M3: the achievement rate increases from 65% to 95%, and the claimed electricity with conflicts is simultaneously reduced to zero. It should be noted that M4 still has one training task that fails to meet the label requirement because of a tight window and large workload. This task has a workload of 1760 units and an execution window of 13 h, and its expected carbon intensity is slightly higher than the committed upper limit of 15 $kgCO_2$ /unit workload. Therefore, it fails to meet the label requirement and falls into Grade C. This indicates that, under given green-electricity resources and execution-window constraints, task-level labels cannot all be achieved without cost.

5.3.2. Task-level carbon traceability and label results of the modified IEEE 118-bus system

As shown in Table 12 and Fig. 8, in the modified IEEE 118-bus system, the two-week facility-scope energy consumption of the 60 label tasks is approximately 3091 MWh. The claimed electricity of M1–M4 corresponds to low-carbon coverage rates of approximately 30%, 46%, 66%, and 82%, respectively, which are at the same level as those in the Chenggong park case. Carbon intensity and coverage still satisfy the

Table 11
Comparison of carbon traceability and label metrics for label-applying tasks in the Chenggong park.

Method	Grade A	Grade B	Grade C	Grade D	Achievement rate/ %	Average carbon intensity*	Low-carbon electricity claimed by tasks/MWh	Claimed electricity with conflicts/MWh
M1	0	3	11	6	15.0	28.2	331.6	0.0
M2	0	5	14	1	25.0	20.7	509.0	31.4
M3	3	10	7	0	65.0	13.1	697.8	41.2
M4	5	14	1	0	95.0	8.4	861.7	0.0

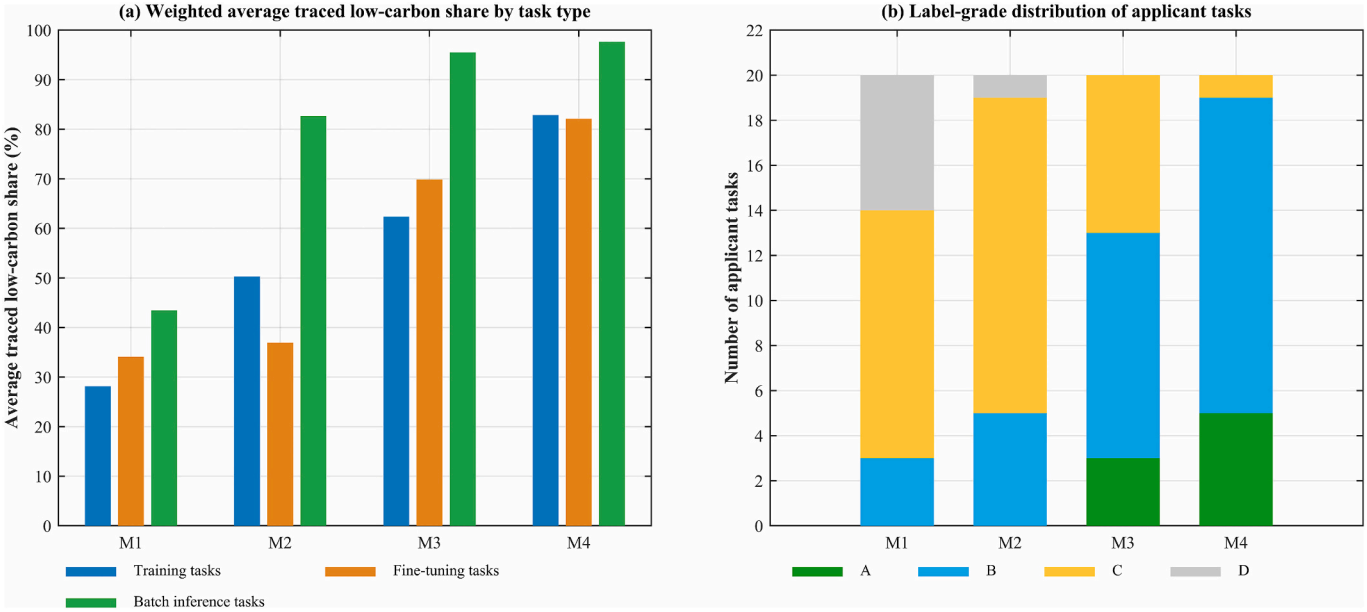


Fig. 7. Task-level carbon traceability and low-carbon label results of the Chenggong park.

Table 12
Comparison of carbon traceability and label metrics for label-applying tasks in the modified IEEE 118-bus system.

Method	Grade A	Grade B	Grade C	Grade D	Achievement rate/ %	Average carbon intensity*	Low-carbon electricity claimed by tasks/MWh	Claimed electricity with conflicts/MWh
M1	0	5	36	19	8.3	27.3	934.1	0.0
M2	0	16	41	3	26.7	21.9	1435.3	92.5
M3	4	32	24	0	60.0	14.0	2032.8	118.6
M4	15	42	3	0	95.0	8.7	2529.2	0.0

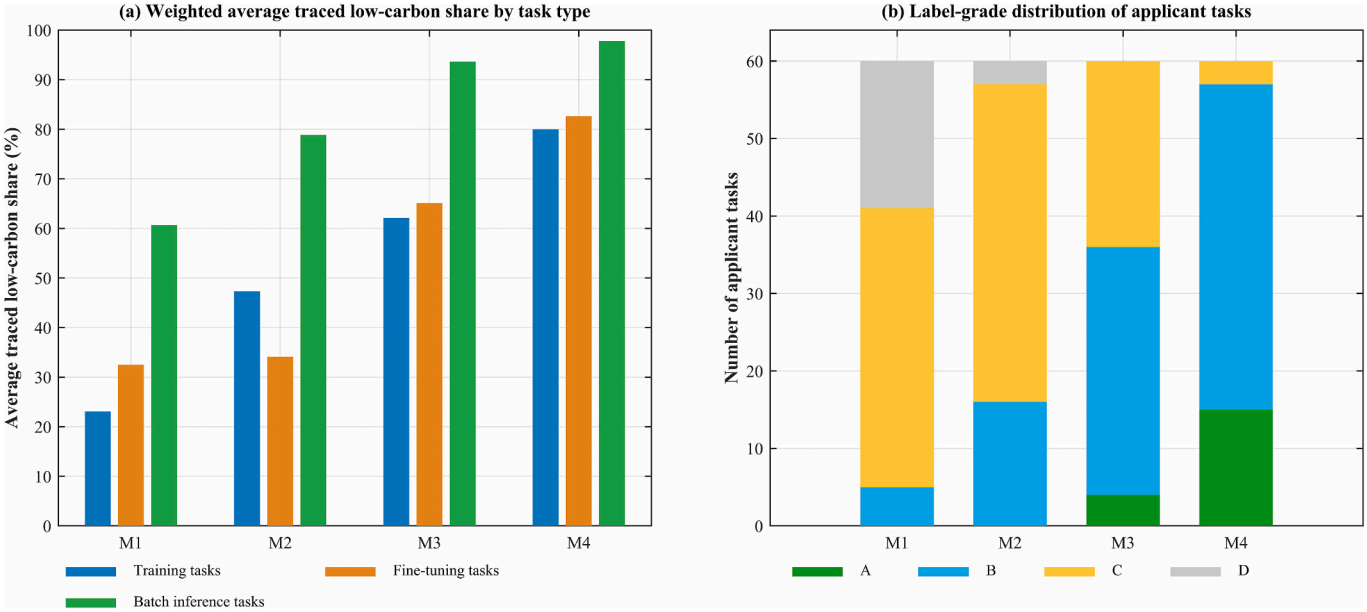


Fig. 8. Task-level carbon traceability and low-carbon label results of the modified IEEE 118-bus system.

above closure relationship, indicating that task-level carbon traceability results remain consistent across different system scales.

In the larger-scale system, the claimed electricity with conflicts in M3 reaches 118.6 MWh, which is significantly higher in absolute value than that in the Chenggong park case. This is because the storage throughput of the three parks is higher, and more green-electricity attributes are transferred across periods through storage. Without a low-carbon attribute inventory mechanism for storage, this part of the claim is difficult to support with verifiable evidence and is therefore more easily removed in ex-post audits. This indicates, from the opposite perspective, that the

more fully storage is utilized, the more necessary the low-carbon attribute inventory mechanism becomes. The label achievement rate of M4 in the modified IEEE 118-bus system is 95.0%, meaning that 57 tasks are achieved and 3 tasks are not achieved. All unachieved tasks are training tasks with tight windows, and no Grade D task appears.

Taken together, the two systems show that the park-level aggregate green-electricity utilization rates of M2–M4 differ only slightly, ranging from 98.6% to 99.3%, whereas the task-level label achievement rate differs by up to 70 percentage points. This indicates that aggregate green-electricity utilization indicators cannot be automatically converted into task-level low-carbon results. Only by simultaneously establishing task-level unique claims, low-carbon attribute inventory in storage, and a label-oriented task-energy coordination mechanism can the label achievement rate and claim credibility be improved on the premise of physically feasible operation and verifiable ledger claims.

5.4. Sensitivity to the residual-mix carbon factor

The residual-mix carbon factor determines the emissions assigned to electricity not covered by a unique low-carbon claim. To examine the influence of this parameter, its scenario-dependent baseline values are uniformly scaled to 80%, 90%, 100%, 110%, and 120% of their original levels. The temporal and scenario-dependent variations are retained, while all task, renewable-generation, electricity-price, cooling, storage, and low-carbon claim parameters remain unchanged. The complete M4 model is reoptimized under each setting for both test systems. The results are presented in Table 13.

The results at the 100% scaling level reproduce the M4 values reported in Tables 9–12. For the Chenggong park, increasing the residual-mix carbon factor from 80% to 120% of its baseline value raises market-based emissions from 2942 to 4132 t CO_2 and the average carbon intensity of label-applying tasks from 7.4 to 9.4 kg CO_2 per workload unit. The label achievement rate decreases from 100% to 90%, corresponding to an increase in the number of unachieved tasks from zero to two. Both unachieved tasks are large training tasks with relatively tight execution windows. Over the same range, the operating cost changes from 3.789 to 3.808 million CNY, a variation of only 0.50%.

The modified IEEE 118-bus system exhibits a similar pattern. Market-based emissions increase from 9495 to 13,424 t CO_2, and the average carbon intensity of the 60 label-applying tasks increases from 7.6 to 9.7 kg CO_2 per workload unit. The number of achieved tasks decreases from 60 to 54, giving a label achievement rate of 90% at the upper sensitivity bound. The operating cost varies by 0.47%, from 12.133 to 12.190 million CNY.

The increase in task carbon intensity is moderated by task shifting, low-carbon electricity allocation, and the intertemporal transfer of low-carbon attributes through storage. These adjustments slightly increase operating cost but limit the deterioration of task-level labels. Across the examined range, both systems maintain label achievement rates of at least 90%, indicating that the main conclusions are not dependent on a single residual-mix carbon-factor setting.

Table 13
Sensitivity of the M4 results to the residual-mix carbon factor.

Test system	Scaling level/%	Operating cost/(10^4) CNY	Market-based emissions/ tCO_2	Average task carbon intensity/ $kgCO_2$ per workload unit	Label achievement rate/%
Chenggong park	80	378.9	2942	7.4	100.0
Chenggong park	90	379.2	3249	8.0	100.0
Chenggong park	100	379.7	3556	8.4	95.0
Chenggong park	110	380.2	3850	8.9	95.0
Chenggong park	120	380.8	4132	9.4	90.0
Modified IEEE 118-bus system	80	1213.3	9495	7.6	100.0
Modified IEEE 118-bus system	90	1214.2	10,528	8.2	98.3
Modified IEEE 118-bus system	100	1215.5	11,530	8.7	95.0
Modified IEEE 118-bus system	110	1217.1	12,495	9.2	93.3
Modified IEEE 118-bus system	120	1219.0	13,424	9.7	90.0

6. Conclusion

This paper addresses the problem of task-level carbon traceability and decarbonization dispatch for AI computing center parks, and constructs a dispatch framework coupling task execution trajectories, park energy-system operation, network-storage auxiliary energy consumption, low-carbon attribute inventory in storage, and task-level low-carbon claims. Under this framework, the carbon label of an AI task is not a simple allocation result of park-average carbon emissions, but is jointly determined by the task execution period, computing resource pool, cooling energy consumption, network-storage energy consumption, storage operation, and unique allocation of low-carbon electricity attributes.

At the model level, this paper embeds task-level carbon traceability into decarbonization dispatch, establishes dispatch constraints consistent with the task ledger, energy-consumption ledger, and low-carbon ledger, and avoids duplicate claims of low-carbon electricity during charging and discharging through the low-carbon energy inventory in storage. Meanwhile, this paper introduces a task-level low-carbon label-violation CVaR constraint to control label tail risk under adverse scenarios such as low renewable output, increased residual carbon factors, and reduced cooling efficiency. The model distinguishes definitional accounting relationships from optimization constraints, and handles issues such as physical closure between network-storage energy consumption and auxiliary power, prohibition of energy and green-electricity allocation to unexecuted tasks, soft terminal constraints of low-carbon inventory, and residual-mix carbon factors. Thus, the proposed method advances decarbonization dispatch of AI computing center parks from aggregate park-level emission reduction to a task-level traceable, verifiable, and risk-controllable dispatch level.

Future work will validate the framework using synchronized task logs, server and facility metering, storage-PCS records, and hourly low-carbon electricity transactions from operating AI computing platforms. Such data will enable component-level energy-error assessment, park-ledger closure tests, verification of unique low-carbon claims, and comparison between model-derived and independently audited task-level carbon labels.

CRediT authorship contribution statement

Yunpeng Jiang: Visualization, Software, Project administration. **Qian Wang:** Writing – review & editing, Writing – original draft. **Guannan Wu:** Writing – review & editing, Writing – original draft. **Tao Huang:** Methodology, Investigation, Funding acquisition, Formal

analysis. **Linhuan Luo:** Writing – review & editing, Writing – original draft. **Qilin Zhou:** Writing – original draft, Visualization, Methodology, Investigation.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Qiang Wang reports financial support was provided by National Natural Science Foundation of China. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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